Ghana

AI-DRIVEN SYSTEM FOR FASHION ANALYSIS FROM LAGOS FASHION WEEK F/W 2025 (HoL – HOUSE OF LASGIDI)

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DECLARATION

I attest that the content of this dissertation report is my own. I certify that the content of this report has not been submitted anywhere else for any award. I also certify that the application's implementation and research have not been submitted elsewhere for any award and are solely owned by me, Anita Dijala. I agree that this work should be saved electronically for academic purposes. I do not give my consent for the work presented in this report, and any and all accompanying documentation, which includes the system implementation, to be made public without ascertaining prior consent and permission from me.

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ABSTRACT

In traditional fashion trend forecasting, there is a large reliance on manual observation and a lack of real-time, data-driven frameworks to quantify choices. Specifically, within the context of African fashion showcases like Lagos Fashion Week, there is a gap in utilizing computational systems to evaluate designer collections. The absence of these datasets makes it technically difficult to train algorithms or accurately replicate the unique physical textures of African fabrics in digital spaces (Boateng et al., 2025). Enforcing these data frameworks is essential for decision-makers who require reliable metrics to improve the production chain and establish sustainable infrastructure across the continent.

To address this, the project introduces 'House of Lasgidi' (Hol), an AI-driven computer vision and sentiment analysis system designed to evaluate the Lagos Fashion Week F/W 2025 show. For colour sentiment analysis, the system evaluates and compares two semantic segmentation pipelines, DeepLabV3+ and FashnHumanParser, followed by K-Means clustering. Textile texture sentiment analysis is achieved by extracting visual features using the Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP), which are classified using Support Vector Machine (SVM). The insights collated are aggregated into a dashboard for brand-specific analytics.

The system testing compared the colour segmentation pipelines, revealing that FashnHumanParser achieved a superior accuracy of 72.73% On the other hand, the GLCM-LBP-SVM pipeline successfully classified fabrics with 91.36% accuracy.

These results demonstrate that HoL provides a scalable, objective tool for designers, buyers, and forecasters to derive actionable, data-driven business insights from runway showcases, effectively bridging the knowledge system gap in African fashion.

1 INTRODUCTION

1.1 Context and Motivation

The necessity for such frameworks is supported by recent findings that for the African fashion industry to progress and thrive, individuals, like designers, buyers, and forecasters, in the fashion industry require more dependable data than is currently available (UNESCO, 2023). Existing industry data is largely incomplete, focusing mainly on raw textile production rather than the high-fashion sector or specific consumer design sentiments (UNESCO, 2023). This lack of analytical infrastructure emphasizes on wider challenges in the region, including a drawback in the infrastructure required to scale creative cooperations (UNESCO, 2023).

Current fabric data in Africa is heavily fragmented and sparse. This data gap is observed by a critical lack of digital archives, as traditional textile knowledge is frequently preserved through oral traditions rather than machine-readable formats (Boateng et al., 2025). As the global fashion industry moves toward the "Fashion 4.0" era, which is defined by big data and artificial intelligence. There is a need to bridge this gap by digitizing and quantifying the visual trends of African runways, specifically within high-growth hubs like Lagos, Nigeria and more.

1.2 Problem Statement

Despite the creative success of events like Lagos Fashion Week (LFW), the analysis of these collections remains a manual, qualitative process. Stakeholders lack automated tools to:

- Objectively quantify design choices and evaluate designer collections using real-time, data-driven frameworks.
- Train algorithms on large-scale datasets specifically focused on traditional textiles, which makes it technically difficult to accurately replicate the unique physical textures of African fabrics in digital spaces.
- Analyze high-fashion or design sentiments, as existing statistical data tends to focus almost exclusively on raw textile and apparel production.
- Track broader consumer trends and behaviours across different geographical regions due to the fragmented and sparse nature of current industry data.

Without a computational solution, the "sentiments" of African design remain unmapped, making it difficult for local designers to compete in a global market that relies on rapid trend data.

1.3 Aims and Objectives

This project aims to develop "House of Lasgidi", an AI-driven web platform that automates the extraction and visualization of fashion trends from Lagos Fashion Week 2025.

To achieve this, the following objectives were established:

- Segmentation: To implement a Semantic Segmentation pipeline to isolate garment pixels from runway images.

- Classification: To design a textile texture classifier using Support Vector Machines (SVM) and statistical features (GLCM + LBP) to categorize textiles into commercial profiles.
- Quantization: To utilize K-Means Clustering in the CIELAB colour space to extract accurate colour palettes.
- Visualization: To build a full-stack dashboard (React + FastAPI) that presents these insights through interactive charts and collection summaries.

1.4 Report Structure

The remainder of this dissertation report is organized as follows:

- Chapter 2: Literature Review. Reviews the evolution of Computer Vision in fashion and explores the mathematical theories behind textile texture and colour analysis.
- Chapter 3: Design. Details the system architecture, the decoupled nature of the API, and the logic of the machine learning pipeline.
- Chapter 4: Implementation. Describes the technical development process, including data scraping, model training, and frontend construction.
- Chapter 5: System Walkthrough. Demonstrates the final "House of Lasgidi" platform and its core functionalities.
- Chapter 6: Testing and Evaluation. Presents a critical analysis of the system's performance, including classifier accuracy.
- Chapter 7: Conclusion. Summarizes the project's achievements and reflects on future work for African fashion informatics.

1.5 Ethical Considerations

The development of the House of Lasgidi system involved several ethical considerations relating to data sourcing, algorithmic fairness, and responsible presentation of AI-generated outputs in a professional creative context.

Data Sourcing and Web Scraping: The runway images used in this project were collected via automated web scraping from publicly accessible fashion websites, primarily BellaNaija Style and the official Lagos Fashion Week domain. All scraped content was used exclusively for academic research and was not redistributed or commercialized. No authentication barriers or paywalls were bypassed during data collection.

Algorithmic Bias and Dataset Fairness: The SVM classifier was trained on a dataset that heavily underrepresents traditional African textiles relative to Western fabrics. This imbalance means the system is technically better equipped to identify Western fabric profiles than traditional African ones. Representational bias is both acknowledged and critiqued throughout this report. This system does not claim to be a definitive authority on African textile classification; rather, it demonstrates the plausibility of the pipeline and explicitly backs up the data gap as a finding in its own right.

Designer and Brand Representation: The system generated AI-generated and AI-driven texture and designer profile outputs for individual designer collections. These outputs are framed explicitly as visual surface texture estimations derived from pixel analysis, not authoritative assessments of a designer's material choices or creative intent. The texture profile report component includes a disclaimer on the dashboard stating that, rather than their fibre composition, ensuring that users interpret the results within their correct technical scope.

Intended Use: The HoL platform is designed as a supplementary analytical tool for fashion professionals, not a replacement for human expertise or editorial judgment. All results presented are derived from a single fashion season's data and should not be interpreted as characterizations of any designer's practice.

2 LITERATURE REVIEW

2.1 Introduction

The intersection of Artificial Intelligence and fashion analytics is a rapidly evolving field. To build an effective system capable of interpreting complex runway data, it is necessary to examine the underlying technologies that enable automated visual analysis. This chapter delineates the problem of cross-domain visual analysis, specifically the "data gap" in African fashion, and evaluates the various solutions proposed in contemporary research.

2.2 The Data Revolution in Fashion

The beginning of the internet and e-commerce has brought about a considerable increase in the amount of visual data available. This exponential increase in volume has been termed "big data". In the context of the global fashion industry, this has sparked a need for systems that extract meaningful insights from millions of unstructured images. Consequently, the discipline of Fashion Informatics has emerged, applying data mining and artificial intelligence to predict trends, optimize supply chains, and understand consumer sentiments.

As stated by UNESCO (2023), there is a severe geographic inequality in this data revolution. While Western fashion ecosystems benefit from massive, well-annotated digital archives, such as the DeepFashion benchmark, African fashion data remains heavily fragmented. Traditional textile knowledge is frequently preserved through oral traditions rather than machine-readable formats (Boateng et al., 2025). This lack of centralized digitized data presents a significant barrier to applying standard Machine Learning algorithms to African runway showcases.

2.3 Computer Vision and Semantic Segmentation in Fashion

Computer Vision aims to develop systems that possess as much visual comprehension as humans. A major subfield of this is identifying and isolating specific objects within an image.

2.3.1 The Evolution from Bounding Boxes to Pixel-Level Masks

Before a machine can analyze the fabric or colour of a garment, it must first locate the garment within an image. In early fashion AI systems, researchers utilized simple bounding boxes (Object Detection). However, bounding boxes capture background noise, and in many legacy datasets, these boxes were "estimated from labelled landmarks, making them noisy" and insufficient for capturing "rich variants of clothing images" (Ge et al., 2019).

Semantic segmentation solves this by classifying each pixel in an image, efficiently filling a "significant gap from real-world scenarios" present in older benchmarks that lacked "per-pixel mask annotations" (Ge et al., 2019). Long et al. (2015), also mentioned in their breakthrough work on Fully Convolutional Networks (FCNs) that semantic segmentation classifies at the image level, creating a precise topological mask of the object.

While general-purpose models, such as Mask R-CNN or DeepLabV3+, have been widely used, they are typically trained on general-purpose datasets (like COCO) and tend to segment the entire human silhouette, grouping skin and hair with clothing. These general predictions are frequently very coarse and not optimal for the required fine-grained analysis in fashion.

2.3.2 Domain-Specific Human Parsing

Recent literature highlights the necessity of domain-specific models to handle the large deformations, occlusions, and discrepancies unique to the fashion industry (Ge et al., 2019). The concept of Fashion Parsing was discovered by Yamaguchi et al. (2012). They demonstrated that models trained specifically on human-clothing topography dramatically outperform general models.

Human parsing networks, such as FashnHumanParser, use Contextualized Convolutional Neural Networks (Co-CNN) to provide granular, multi-class segmentation (separating upper garments, pants, skirts, and dresses) by predicting dense landmarks and per-pixel masks for specific categories (Ge et al., 2019). Utilizing a specialized parser ensures that the following colour and texture algorithms process only "garment pixels", as these domain-specific representations capture the unique shapes and structures of clothes, which are significantly different from human pose (Ge et al., 2019). The HoL system can aggressively subtract skin and background noise, ensuring that only purified garment pixels are passed to the analytical engines.

2.4 Texture Analysis Approaches: Deep Learning vs. Handcrafted Features

Classifying textile types from 2D images is a complex challenge known as texture analysis. In computer vision, texture analysis is the mathematical quantification of the spatial arrangement of colour or intensities. The current academic landscape is largely divided into two paradigms: Convolutional Neural Networks (CNNs) and Handcrafted Statistical Features.

2.4.1 Convolutional Neural Networks (CNNs)

Models like ResNet or VGG16 automatically learn feature representations from raw pixels, but the process of estimating millions of parameters means they require a very large number of annotated images for successful convergence (Liu et al., 2019). When applied to smaller, highly specialized domains, such as identifying unique African textiles, deep learning techniques have been found to be inadequate because they struggle with fine-grained inter-class appearance variations that define specific materials (Hu et. Al., 2020). CNNs are highly prone to overfitting, memorizing the training images rather than learning the underlying textural patterns.

Consequently, in situations with limited training data, CNNs act as "black boxes" that are highly prone to overfitting and are significantly outperformed by handcrafted features like Local Binary Patterns (LBP) (Liu et al., 2019). Such descriptors provide essential rotation and gray scale invariance while maintaining low computational complexity (Ojala et al., 2002; Liu et al., 2019).

2.4.2 Handcrafted Statistical Features (GLCM and LBP)

Alternatively, statistical approaches manually extract mathematical properties from an image.

- Macro-Texture (GLCM): The Gray-Level Co-occurrence Matrix (GLCM), was first pioneered by Harlick (1973), evaluates the macro-spatial relationship between pixels. By analyzing a grayscale matrix across multiple orientations, GLCM extracts rotation-invariant metrics such as Contrast, Energy, and Homogeneity.
- Micro-Texture (LBP): To complement this, Local Binary Patterns (LBP) analyze micro-textures by thresholding neighbouring pixels, providing a highly effective measure of surface roughness (Ojala et al., 2002).

Kumar (2008) asserts that fusing GLCM and LBP descriptors results in highly accurate, computationally lightweight feature vectors. When these features are passed into a Support Vector Machine (SVM), the classifier achieves high generalization without the overfitting risks of deep neural networks.

2.5 Colour Quantization and Perception

Extracting a dominant colour palette from clothing requires unsupervised learning, as the colours are unknown before analysis. K-Means clustering is the industry standard for this task, effectively grouping millions of pixels into a predefined number of spatial centroids (k).

However, the mathematical space in which this clustering occurs is critical. Digital images are natively stored in the RGB colour space. Literature in human-computer interaction stresses that RGB is not “perceptually uniform” (Plataniotis & Venetsanopoulos, 2000). The mathematical distance between two RGB values does not correlate with how the human eye perceives the difference between those colours. To resolve this. Modern fashion retrieval systems transform RGB arrays into the CIELAB (LAB) colour space before clustering. Because the LAB space mimics human vision, the K-Means algorithm produces commercial palettes that are highly accurate to the human observer.

2.6 Existing Systems and the Cross-Domain Gap

To situate the House of Lasgidi (HoL) system within the broader technological landscape, it is necessary to critically examine the existing commercial and academic systems that currently define fashion trend forecasting and garment analysis. While the field has advanced considerably, a systematic review reveals a persistent and significant cross-domain gap when these tools are applied to non-Western, and particularly African fashion contexts.

2.6.1 Commercial Forecasting Platforms

WGSN is the industry’s dominant trend forecasting platform, used by major global fashion brands and retailers to predict colour cycles, silhouette trends, and consumer sentiment season in advance. It combines editorial expertise and data aggregation from social media, search engines, and retail performance metrics. However, the platform’s analytical pipeline is entirely exclusive, making independent academic validation of its underlying algorithms impossible (Amed et al., 2019). WSGN's data sourcing is focused on Western fashion markets.

Heuritech is a platform employing deep learning for fashion trend forecasting. The system uses Convolutional Neural Networks (CNNs) to analyze millions of social media images, automatically identifying and tracking the frequency of specific garment attributes, such as colour, silhouette, and prints. The system's training data source is almost exclusively from Western social media platforms and fashion week imagery. As a consequence, its classifiers are not equipped to recognize traditional African textiles, which do not appear in sufficient volume within its source data to generate meaningful predictions.

These platforms collectively represent a well-resourced technological ecosystem that, by design or by omission, excludes the African fashion segment from its analytical scope.

2.6.2 Prior Computational Frameworks

While commercial platforms remain geographically biased, several academic studies have found that machine learning is being used for traditional and specialized textile analysis. These works provide technical validation for the architectural choices in the HoL system.

- Recognition of Traditional African Patterns (Olawale & Ajayi, 2013): One of the efforts to distinguish African textiles used a two-level classification scheme to identify designs such as Adire, Kente, and ASO-Oke. Utilizing Principal Component Analysis (PCA) and Discrete Wavelet Transforms (DWT) for dimensionality reduction, researchers employed artificial neural networks to achieve reliable recognition. This study establishes the technical ability of using computational models to manage the complex geometric and aesthetic features that define African fabrics. HoL integrates human parsing, texture classification, and colour sentiment analysis into an aggregated dashboard designed specifically for business insights.
- Colour Palette Extraction from Runways (Lai & Westland, 2020): The HoL system's colour analysis approach is supported by research demonstrating that image segmentation before clustering significantly improves accuracy.
- Texture Analysis on Microscopic Surfaces (Seçkin et al., 2023; Koval, 2018): The effectiveness of combining handcrafted features like GLCM and LBP is further validated by the FabricNET project and studies on Zulu fabrics. HoL operates on runway imagery, introducing challenges like lighting artefacts and model poses.

2.6.3 Academic Fashion AI Systems and Benchmark Datasets

The DeepFashion dataset (Liu et al., 2016) is the foundational benchmark for most academic fashion AI research. Containing over 800,000 annotated clothing images, it has enabled significant advances in garment retreval, attribute recognition, and cross-domain alignment between runway and consumer imagery. However, the dataset is sourced entirely from Western fashion retail and editorial imagery. As Boateng et al. (2025) noted, this geographical bias creates a structural barrier: algorithms trained on DeepFashion have no learned representation of the unique textural and structural properties of traditional African fabrics, severely limiting their cross-domain applicability.

FashionNet (Liu et al., 2016) is a deep learning framework that predicts clothing attributes and body landmark locations from a single image. By learning a shared feature representation for both tasks, FashionNet demonstrated that contextual spatial information, like knowing where a collar or hem is located, significantly improves attribute classification accuracy. This architectural insight informs the HoL system's use of human parsing as a prerequisite step before colour and texture analysis. However, like DeepFashion, FashionNet was trained and evaluated entirely on Western retail imagery, establishing a target that does not transfer to the distinct visual domain of African runway fashion.

2.6.4 The Cross-Domain Gap and the HoL Contribution

The review of these systems reveals a consistent and compounding cross-domain gap. First, commercial platforms operate on proprietary architectures trained on Western data, making their outputs largely irrelevant to African fashion designers and buyers. Second, even the most advanced open academic datasets, DeepFashion and FashionNet, are anchored in Western retail imagery and carry no learned representation of traditional African textiles. As UNESCO (2023) affirms, the absence of these analytical frameworks is a critical barrier to the growth of the African fashion sector, as stakeholders lack the data infrastructure needed to make evidence-based decisions.

The House of Lasgidi system directly addresses this gap. Rather than attempting to fine-tune a model pre-trained on a Western dataset, the HoL pipeline is deliberately constructed around handcrafted features (GLCM + LBP), which do not require large pre-annotated datasets to generalize effectively. This design decision allows the fabric analysis engine to be trained on a curated dataset of some traditional African textiles sourced from publicly available African Fabric datasets (Mikuns, 2021). This approach demonstrates that a decoupled, pipeline-based AI architecture can successfully interpret non-Western runway data without the prohibitively large training sets that deep learning alternatives demand, offering a scalable and reproducible blueprint for other African fashion week showcases.

3 DESIGN ANALYSIS

3.1 Introduction

This chapter discusses the design of major components and details of the methods used for the House of Lasgidi (HoL) framework. Emphasizing on the limitations identified within the fashion data landscape in Nigeria, this design presented here focuses on an AI-driven pipeline capable of objective quantification. The design follows a modular architecture, separating the heavy computational tasks of garment segmentation and texture feature extraction from the user-based visualization dashboard. This approach was chosen to enable the comparative testing of colour pipelines while maintaining a seamless user experience for fashion professionals. Furthermore, this section illustrates the functional and non-functional requirements of the system.

3.2 Requirements Specification

The design of all systems follows a basic structure of two major requirements, namely, functional and non-functional requirements. Therefore, the system specification is broken down into two parts. The functional requirements of this system focus on immediate concerns as opposed to the non-functional requirements; therefore, this section of the report will give a detailed description and illustration of the functional and non-functional requirements for HoL.

3.2.1 Functional Requirements

The functional requirements define the specific behaviours and services the system must provide:

- FR1 (Data Acquisition): The system shall use a Selenium-based web scraper to automatically collect raw runway images from designated fashion showcase web sources.
- FR2 (User Authentication): The system shall allow users (Designers, Buyers, and Students) to create accounts, log in, and log out to access restricted brand analytics.
- FR3 (Image Pre-processing): The system shall perform image normalization and resizing to prepare raw data for the segmentation pipelines.
- FR4 (Garment Segmentation): The system shall implement segmentation using DeepLabV3+ and FashnHumanParser to segment fashion garments from runway backgrounds.
- FR5 (Colour Sentiment Analysis): The system shall extract dominant garment colours and perform K-Means clustering to generate colour palettes and frequency data.
- FR6 (Fabric Sentiment Analysis): The system shall extract texture features using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP) for classification via a Support Vector Machine (SVM).
- FR7 (Brand-Specific Filtering): The system shall allow users to select specific fashion brands to view expected collection analysis.

- FR8 (Interactive Visualization): The system shall present analysis results via a dashboard featuring colour frequency charts, fabric frequency charts, colour palette displays, and ranked trend lists.
- FR9 (Insight Reporting): The system shall generate and display "Collection Insight Reports" that summarize the overall colour and fabric trends of a specific designer's show.

3.2.2 Non-Functional Requirements

Non-functional requirements specify the criteria used to judge the operation of the system rather than specific behaviours.

- NFR1 (Predictive Accuracy): The fabric classification (SVM) and colour segmentation pipelines must maintain a high accuracy threshold (above 70%) to provide trustworthy data for fashion professionals.
- NFR2 (Usability): The dashboard interface must be designed for non-technical fashion professionals, ensuring that trend insights are interpretable without requiring data science expertise.
- NFR3 (Scalability): The system's "House of Lasgidi" framework should be capable of scaling to handle increasing volumes of image data from various fashion weeks.
- NFR4 (Robustness): The computer vision modules must be robust enough to handle varied runway conditions, including different lighting and complex fabric textures.

3.3 Use-Case Model

3.3.1 Actors

This system has three main actors, each representing a distinct stakeholder in the fashion industry:

- Designers - a person who views and filters all the analysed results, insights, and only views the fabric-to-colour compatibility insights.
- Buyers – a person who views and filters all the analysed results, insights, and only views the commercial insights.
- Students – a person who views and filters all the analysed results and insights.

3.3.2 Use-Case Diagram

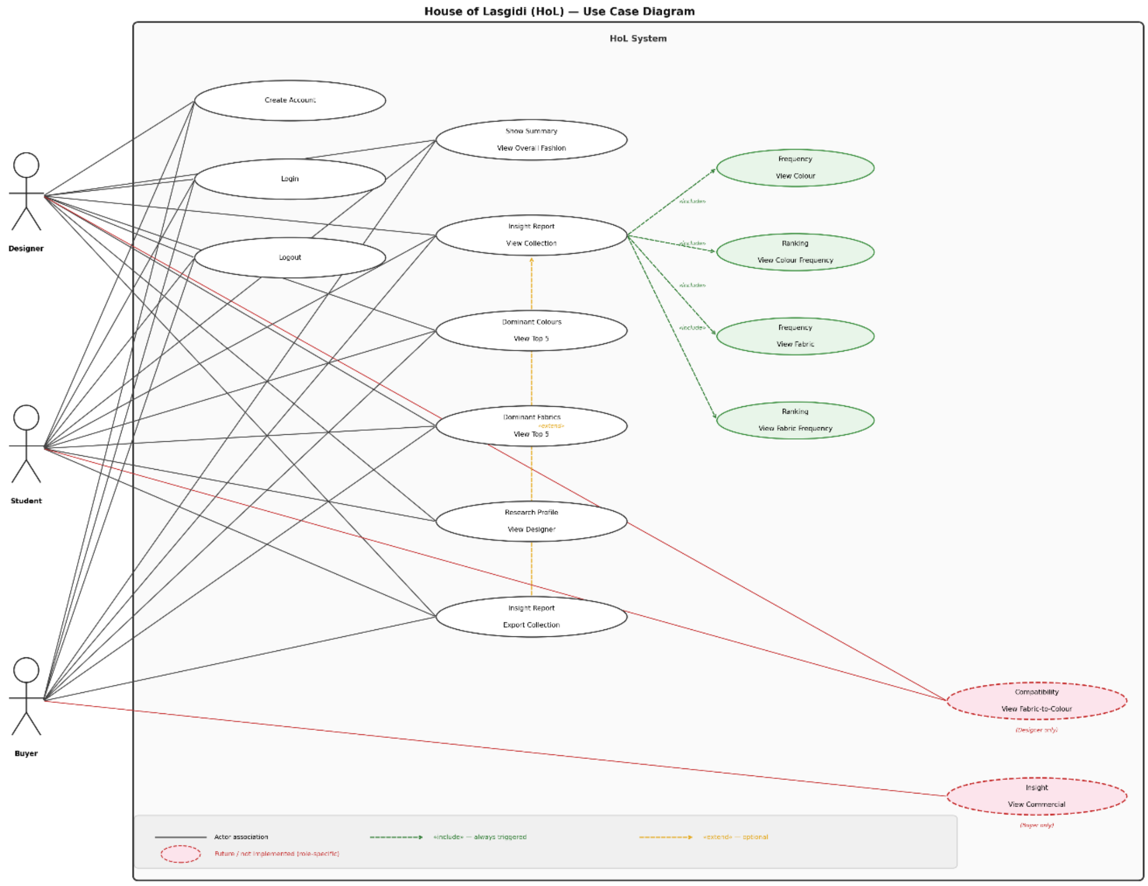

Figure 1: House of Lasgidi (HoL) Use Case Diagram

3.3.3 Use-Case Scenarios

Table 1: Use-Case Scenarios Table

ID	Use Case	Prerequisites	Use Case Description	Actor(s)	Success Scenario
UC-1	Create Account	The actor is not currently logged into the systems and possesses a valid email address.	Allows a new individual to register and set up a profile within the HoL system.	Student, Buyer, Designer	The actor navigates the registration page, inputs their details and submits. The system validates the data and creates a new profile in the database, and redirects the actor to the login page.

UC-2	Login	The actor has registered an account on the system	Authenticates the user's credentials to grant access to the system's dashboard and features.	Student, Buyer, Designer	The actor inputs registered details and the system authenticates, initiates a secure session, and redirects the actor to the main HoL dashboard.
UC-3	Logout	The actor is currently logged into the system with an active session.	Securely terminates the user's session in the system.	Student, Buyer, Designer	The actor clicks the 'Logout' button. The system securely terminates the active session and redirects the actor to the public landing page.
UC-4	View Overall Fashion Show Summary	The actor is logged in. The JSON file containing the show analysis results exists.	Displays a macro-level dashboard summarizing the trend data for the entire Lagos Fashion Week showcase.	Student, Buyer, Designer	The system displays the Overview section and renders the charts.
UC-5	View Collection Insights Report	The actor is logged in. The JSON file containing the show analysis results exists.	Presents a detailed brand-specific analytics report, pulling in data such as colour and fabric frequency for a specific designer's collection.	Student, Buyer, Designer	The actor selects a brand, and the system renders the analysis dashboard.
UC-6	View Top 5 Dominant Colours	The actor is logged in. Brand or overview summary is active. The	Visualizes the five most frequently occurring colour palettes detected	Student, Buyer, Designer	The system extracts the top 5 colours and percentages from the

		JSON file contains frequency counts	by the AI in the runway images.		JSON file and renders them as a palette.
UC-7	View Top 5 Dominant Fabrics	The actor is logged in. Brand or overview summary is active. The JSON file contains SVM classification results.	Visualizes the five most frequently used fabrics detected by the AI in the runway images.	Student, Buyer, Designer	The system extracts the data from the JSON file and renders it as a bar chart.
UC-8	View Fabric Frequency Ranking	The actor is logged in. Brand or overview summary is active.	Displays a ranked list of the top 5 dominant fabrics identified	Student, Buyer, Designer	The system extracts the fabric classifications from the JSON file and displays them in a ranked list.
UC-9	View Colour Frequency Ranking	The actor is logged in. Brand or overview summary is active. The JSON file contains frequency counts	Displays a ranked list of the top 5 dominant fabrics identified	Student, Buyer, Designer	The system extracts the top 5 colours and percentages from the JSON file and renders them in a ranked list.
UC-10	View Designer Research Profile	The actor is logged in. The designer JSON file exists.	Displays background information regarding a specific designer.	Student, Buyer, Designer	The system retrieves the specific designer's bio and background data and displays it on the profile view.
UC-11	View Commercial Insights	The actor is logged in.	Provides actionable business intelligence and market trend predictions from the runway data	Buyer	The system displays actionable business insight.

UC-12	Export Collection Insight Report	The actor is logged in. The specific brand report is displayed in the dashboard.	Allows the user to download the generated analytics report for offline use or presentations.	Student, Buyer, Designer	The system compiles the data currently in the dashboard view into a PDF file for the user to download.
UC-13	View Fabric-to-Colour Compatibility	The actor is logged in. The actor has a 'Designer' role.	Provides advanced insights into how specific runway fabrics are paired with specific colour sentiments.	Designer	The system displays a compatibility matrix showing which fabrics were most frequently paired with specific colour sentiments.

3.4 Architectural Design

3.4.1 Overall Architecture

The architecture of House of Lasgidi (HoL) follows a structured pipeline-oriented design. This system is divided into five distinct layers, illustrated in Figure 2.

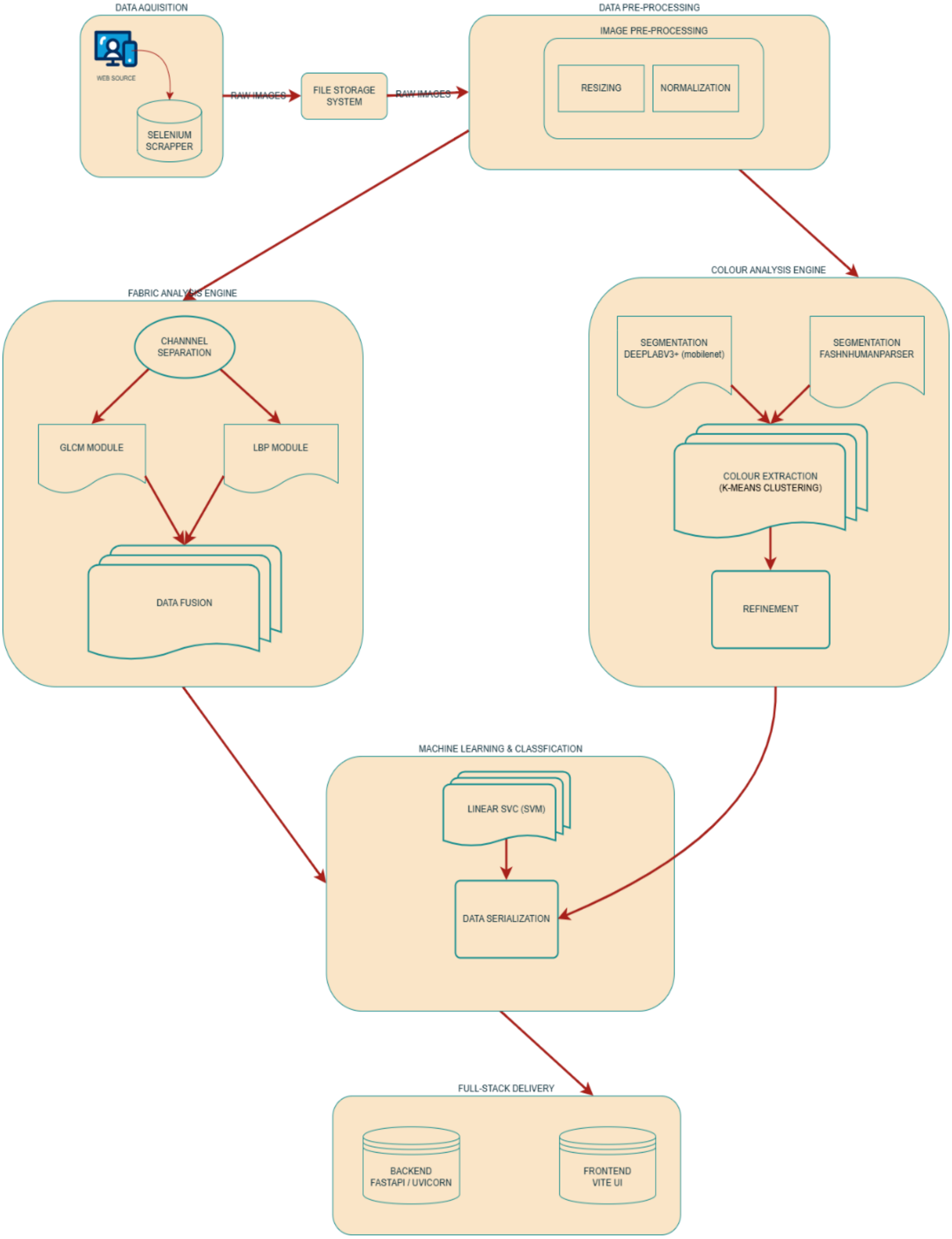

Figure 2: House of Lasgidi (HoL) Overall Architecture Diagram

The five layers are:

1. Data Acquisition Layer: Responsible for sourcing raw runway imagery via automated web scraping.
2. Data Pre-Processing Layer: Normalizes, resizes, and converts all images into a consistent format before analysis.
3. Feature Extraction Layer: Contains two parallel analytical engines, Colour Analysis Engine (comparative segmentation between DeepLabV3+ and FashnHumanParser followed by K-Means clustering) and the Fabric Analysis Engine (GLCM + LBP feature extraction).
4. Machine Learning and Classification Layer: Deals with the Linear SVM classifier and the data serialization pipeline that converts ML outputs into structured JSON.
5. Full-Stack Delivery Layer: Delivers the serialized results to the user via a FastAPI backend and a React frontend dashboard.

This architecture is suitable for this system because it allows the image processing to stay separate and not interfere with the responsiveness of the user dashboard. This also allows for the overall system to be easily understood at each layer of the system. The details of this structure and the role of each of these layers will be discussed in more detail.

3.5 Module Decomposition

3.5.1 Data Acquisition & Image Pre-Processing Module

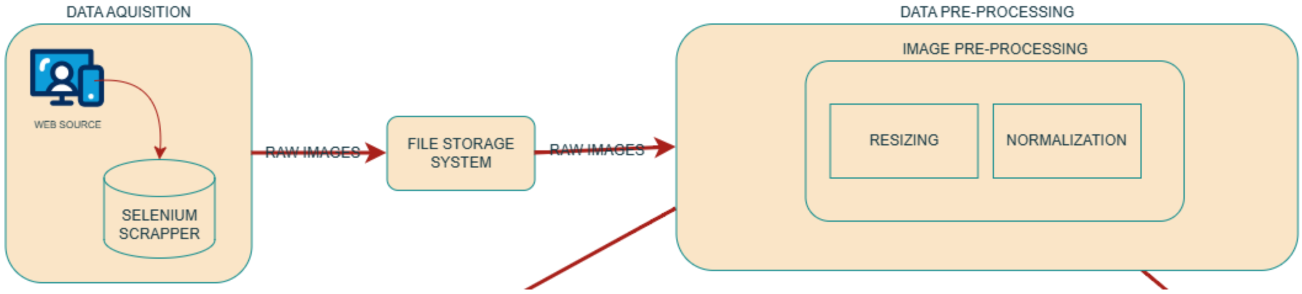

Figure 3: Data Acquisition & Pre-Processing Modules Architecture

This module is responsible for carrying out acquiring the raw runway image dataset and preparing it for analysis.

Beginning with web scraping, the use of Selenium to scrape and download the runway images from fashion blogs. In regards to data cleaning, the majority of the dataset was acquired using Selenium. The system encountered structural inconsistencies on the official Lagos Fashion Week domain that bypassed automated logic, as the website is a dynamic page. Traditional scrapers that only use low-level HTTP requests may fail because they cannot access dynamic material created by client-side scripts. To ensure a complete dataset for the F/W 2025 showcase, a hybrid approach was adopted, where the automated scraping was replaced by manual data retrieval for three specific brands.

To maintain the integrity of the sentiment analysis, the raw scraped data underwent manual cleaning. This prevents unpredictable consequences associated with the GIGO (Garbage in, garbage out) principle (Wang and Wang, 2019). Irrelevant images are seen as noise and can introduce outliers into the downstream analytical engines.

In the Data Pre-Processing module, all the images were normalized to a fixed resolution to ensure consistent input dimensions across the pipeline. The images were then converted to a standardized colour space to prevent errors in imaging (Inoue and Yagi, 2020). Finally, the images were converted into normalized floating-point tensors, so the images could be used for segmentation neural networks.

3.5.2 Segmentation Module

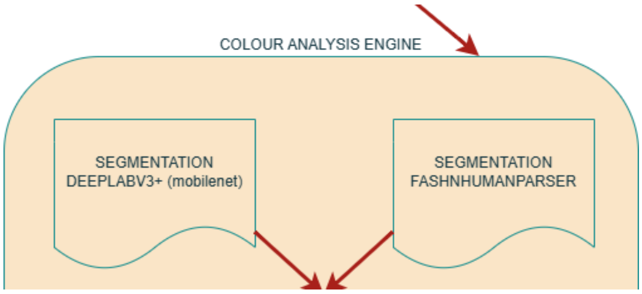

Figure 4: Segmentation Module Architecture Diagram

This module is responsible for isolating the models in the runway images from complex background elements, as it is an important prerequisite for the Colour Analysis Engine. If the background, skin, or hair is not filtered out, colour clustering algorithms will process heavily skewed data.

This segmentation module extracts precise garment masks using two distinct convolutional neural network (CNN) pipelines to determine the optimal approach for runway imagery: the DeepLabV3+ pipeline and the FashnHumanParser pipeline.

DeepLabV3+ Pipeline: The first pipeline uses the DeepLabV3+ model pre-trained on COCO dataset. The model performs pixel-wise classification, a standard semantic segmentation task with the goal of assigning labels to "every pixel in an image" (Chen et al., 2018). While this model is highly efficient due to its use of depth-wise separable convolution to drastically reduce complexity (Chen et al., 2018), this general-purpose model presented challenges for the analysis. The model segments the entire human silhouette, including skin, hair, and accessories, which introduces noise into the colour analysis. Furthermore, the model demonstrates a difficulty in segmenting heavily occluded objects, which is common in runway environments (Chen et al., 2018).

FashnHumanParser Pipeline: A second pipeline was integrated using the FashnHumanParser, a domain-specific human parsing network which leverages a specialized Contextualized Convolutional Neural Network (Co-CNN) designed to decompose human images into distinct semantic clothing and body regions (Liang et al., 2015). The system is programmed to aggressively filter out non-garment pixels by exclusively targeting specific clothing IDs: 4 (upper-clothes), 5 (skirt), 6 (pants), and 7 (dress). These categories are identified by the researchers as essential for clothing style recognition and retrieval (Liang et al., 2015). The output masks are cross-referenced with a predefined LAB-space skin-tone array.

3.5.3 Feature Extraction Module (Textile Analysis)

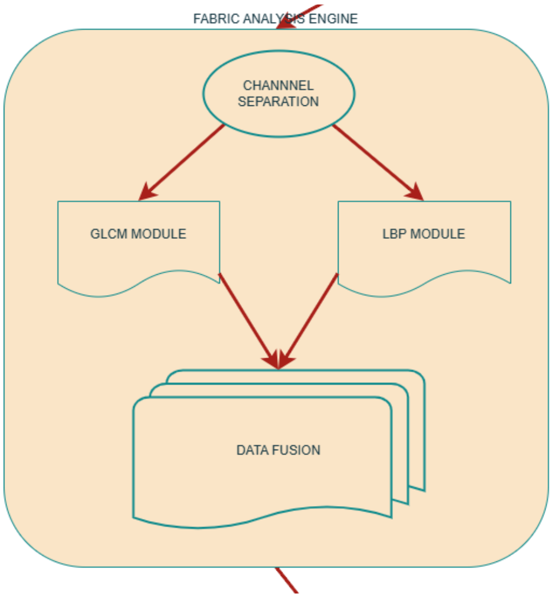

Figure 5: Feature Extraction Module Architecture

Following the data acquisition and manual curation phase, the system executes a specialized feature extraction pipeline to convert the visual properties of the runway images into a quantifiable mathematical format. The fabric analysis engine operates directly on the raw images. This approach ensures that the global textural context and the interplay of light on the fabric's surface are preserved.

The system utilizes a combination of spatial and statistical descriptors to build a robust fabric signature:

Statistical Texture Analysis (GLCM): This system employs the Gray-Level Co-occurrence Matrix (GLCM) to analyze the macro-texture and spatial relationships of pixels within the image. The GLCM evaluates texture by tabulating the frequency at which pairs of specific gray-level values and spatial relationships occur (Haralick et al., 1973). The system extracts four key statistical properties from the matrix that mathematically describe the coarseness and regularity of the fabric weave: contrast, dissimilarity, homogeneity, and energy.

Micro-Pattern Analysis (LBP): To complement the macro-spatial analysis of the GLCM, the system utilizes Local Binary Patterns (LBP) to capture small and fine details, like the distinct weave of Aso-Oke (dense, ridged weave), that statistical methods might miss (Liu et al., 2019). The LBP operator labels the pixels of an image by thresholding the neighbourhood of each pixel and considers the result as a binary number (Ojala et al., 2002). A significant technical addition is the calculation of Shannon Entropy from the LBP histogram result. This provides a high-level discriminator that helps the SVM distinguish between highly patterned traditional textiles and uniform synthetic fibres.

The final stage of this module is the unification of the texture vector. This fused vector represents the unique mathematical identity of the designer's fabric choice. To ensure a decoupled architecture, these features are saved in structured CSV files. This data persistence allows the classification module to be trained and executed without re-processing the raw images, significantly optimizing system performance.

3.5.4 Classification and Clustering Module

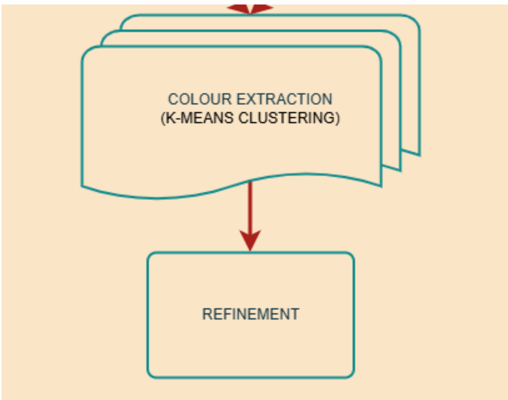

Figure 6: Clustering Module Architecture

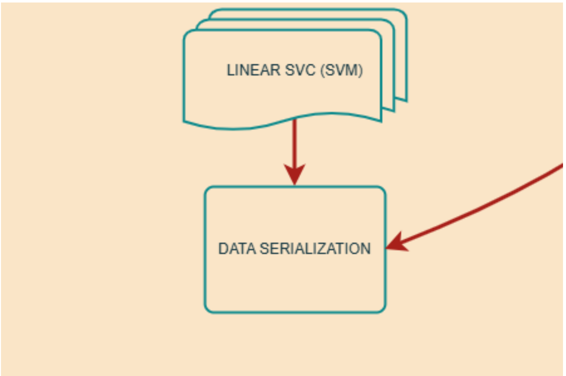

Figure 7: Classification Module Architecture

This module is responsible for the analytical transformation of the mathematical features extracted in the preceding modules into actionable insights. Because the system analyzes two distinct domains, colour and fabric texture, it uses a bifurcated classification and clustering architecture.

K-Means Clustering: To extract the dominant colour palettes from the garments, the system relies on an unsupervised clustering approach. The isolated garment pixels, the output of the FashnHumanParser mask, are converted from the RGB colour space into the CIELAB (LAB) colour space. This conversion is critical because of the Euclidean distances in LAB space closely mimic human visual perception, preventing distorted colour associations (Al-Rawi & Beel, 2020). To filter out the noise, the algorithm enforces a strict weight threshold: only colours that constitute more than 15% of the garment's total surface area are recorded. This ensures that minor accessories or lighting artefacts do not skew the final palette (Karthick et al., 2023).

Support Vector Machine: To classify the complex textural signatures of the GLCM-LBP feature vectors produced by the Feature Extraction Module, the system uses a supervised machine-learning approach. The system deploys a Linear Support Vector Classifier. Given the size of the training dataset, a Linear SVM was chosen over CNN-based alternatives because it achieves high generalization and an accuracy of 91.36%, while mitigating the overfitting risks associated with deep learning on a smaller dataset (Cortes & Vapnik, 1995). Feature normalization is applied before classification to ensure that features of different scales do not disproportionately influence the decision boundary.

3.5.5 Visualisation Module

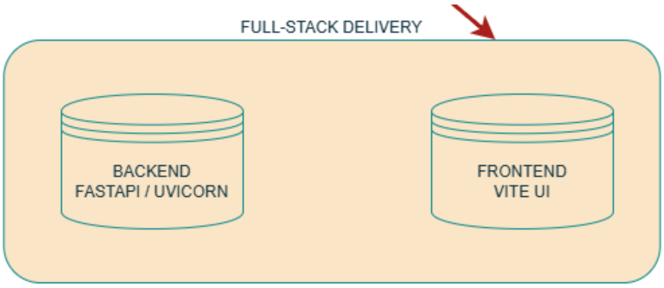

Figure 8: Visualization Module Architecture

The final component of the analytical pipeline is the Visualization Module. The objective of this module is not to generate raw data, but to transform the machine learning outputs of the Classification and Clustering Module into visual representations. Because the target end-users are fashion buyers, students, and designers, the cognitive load of interpreting raw arrays must be minimised.

The module is designed to dynamically render two primary types of visual feedback:

1. Colour Charts: Using the centroid data generated by the K-Means clustering algorithm, the system draws proportional colour blocks to represent the visual weight of each hue within a collection.
2. Fabric Charts: The categorial outputs from the Support Vector Machine (SVM) are translated into percentage-based visualizations to instantly communicate the textural composition of a brand's runway presentation.

3.6 UI/UX Design

The User Interface (UI) and User Experience (UX) of the House of Lasgidi (HoL) system are explicitly designed around the principles of rapid insight retrieval. Fashion industry professionals operate in fast-paced environments where prolonged data interpretation is a bottleneck. Therefore, the interface adopts a minimalist, data-forward design paradigm, prioritizing visual hierarchy over decorative elements (Shneiderman & Plaisant, 2016).

3.6.1 Dashboard Layout & Navigation

To ensure a user-focused experience, the dashboard architecture follows a "Macro-to-Micro" navigational flow. Users are first presented with the global state of the runway before finding the granular data.

The first page of the dashboard acts as the executive summary. It features aggregated statistics for the entire Lagos Fashion Week 2025 event, displaying the most dominant colour across all designers and the overall fabric sentiment. The second page of the dashboard displays the brand-specific analysis. The interface uses a dropdown layout, allowing users to select individual designers. Upon selection, the UI dynamically injects the brand's specific colour insights and fabric insights into the viewport.

The system also includes a dedicated designer profile, which provides qualitative context (biographies, brand ethos, etc.) that complements the ML results, and a PDF export engine that compiles the segmentation masks, K-Means palettes, and designer metadata into a professional offline report.

The HoL Dashboard was designed using a modular, component-based architecture to facilitate both macro-trend analysis and micro-brand validation. The UI layouts for the Overview and Brand Analysis pages are documented in Appendix G and Appendix H, respectively. These layouts demonstrate the implementation of a 'Dashboard-First' philosophy, where complex ML results (K-Means and SVM) are interpreted into interactive, readable visualizations.

3.7 Data Design

The system's scalability and the minimizing of the latency on the front-end of the dashboard are enforced by employing a decoupled data design. Rather than executing live machine learning predictions whenever a user clicks a brand, the system relies on a serialized data bridge.

The output from the colour and fabric engines is structured, mapped to human-readable sentiments, and stored in a hierarchical JSON file. JSON (JavaScript Object Notation) was selected over CSV for this specific layer because its nested key-value structure perfectly mirrors the object-oriented needs of a web dashboard.

While the machine learning outputs are serialized into JSON structure, the system requires a strong delivery mechanism to serve this data to the end-user. The data design employs a stateless RESTful API architecture.

The backend acts as the data provider, exposing specific endpoints to securely transmit the JSON payloads. The dashboard's frontend requests this data asynchronously. This decoupled design ensures that the heavy Python machine-learning environment is completely isolated from the user interface, allowing the dashboard to remain highly responsive. Furthermore, the inclusion of an authentication layer ensures that proprietary fashion analytics remain secure and accessible only to authorized industry professionals.

3.8 Design justification

The architectural and algorithmic choices for House of Lasgidi (HoL) were driven by the specific computational challenges of analyzing complex runway images. While individual module decisions are detailed in Section 3.5, the overarching system design is justified by the following core technical trade-offs:

- FashnHumanParser is chosen over DeepLabV3+: General-purpose segmentation models like DeepLabV3+ segment the entire human silhouette, including skin, hair, and accessories. FashnHumanParser was explicitly chosen because its 18-class schema specifically targets human-clothing topography, allowing the system to aggressively filter out non-garment pixels and ensuring the downstream colour analysis is highly purified.
- GLCM + LBP over Convolutional Neural Networks (CNNs): While CNNs are the modern standard for general image classification, some gaps require massive datasets and are highly prone to overfitting on smaller, specialized datasets available for African textiles. By fusing GLCM and LBP, the system achieves a highly interpretable, lightweight feature vector that is rotation-invariant that generalizes effectively on limited training data, directly addressing the dataset constrain identified in Section 2.6.3.
- CIELAB Colour Space: Raw images are natively processed in RGB. However, the system converts these to the LAB colour space before clustering. This is mathematically crucial because LAB is "perceptually uniform", meaning the Euclidean distance between two points in the LAB space closely matches how the human eye perceives colour differences, preventing the system from making inaccurate colour mappings (Al-Rawi & Beel, 2020).
- Linear Support Vector Machine (SVM) Classifier: SVMs excel at finding the optimal hyperplane in high-dimensional feature spaces, delivering high accuracy and generalization with significantly lower computational overhead compared to deploying a secondary deep learning network on linearly seprarble texture classes (Cortes & Vapnik, 1995).
- FastAPI Backend: FastAPI is built on modern asynchronous Python capabilities and natively validates JSON data types. Most importantly, it seamlessly integrates with the data science libraries (Pandas, Scikit-Learn) used in the analytical pipeline, minimizing friction between the machine learning models and the web server.
- React Frontend: The fashion data being visualized (colours, fabric percentages, trends, etc.) is highly dynamic and requires frequent DOM updates as the user filters between designers. React's component-based architecture and Virtual DOM allow complex visualizations, like the ColourCharts, to re-render instantly without requiring full page reloads, providing a smooth, app-like user experience.

3.9 Constraints

The design and execution of this system operated within several technical and data-based constraints.

Dynamic web scraping: Many official fashion websites employ dynamic JavaScript rendering and strict anti-scraping protocols. Automated web scraping experienced severe latency and required hybrid manual curation for a subset of brands.

African textile data scarcity: There is a distinct lack of large-scale, publicly available datasets specifically focused on indigenous African textiles. To train SVM effectively, the raw runway images had to be merged with a dataset containing low-resolution African fabric images (Mikuns, 2021) and a general Western fabric dataset (Orchit, 2022). The implications of this constraint on classifier performance are discussed in full in Section 6.5.

Stage lighting and colour constancy: Runway imagery is subject to extreme environmental noise, specifically dynamic and coloured stage lighting. Extreme lighting gels can temporarily shift the perceived CIELAB colour values of a garment, meaning the K-Means clustering algorithm may occasionally misclassify a heavily shadowed or tinted fabric, despite the 15% noise-reduction threshold.

Fixed K parameter: The K-Means clustering algorithm used for colour extraction is parametric, meaning it requires a predefined number of clusters, which was set to 8 in this system. While this is highly effective for extracting dominant runway palettes, this mathematical constraint means that minor, highly intricate accent colours on complex garments (like Aso-Oke) may be mathematically swallowed or merged into larger neighbouring centroids.

3.10 Conclusion

The design of the House of Lasgidi system provides a strong, decoupled framework that bridges the gap between complex computer vision algorithms and the fashion industry. The choice of interpretable algorithms over deep learning directly addresses the African textile data scarcity problem. By fusing semantic segmentation, texture analysis (GLCM+LBP), optimized data serialisation (JSON), and RESTful API, the architecture successfully translates raw, unstructured photos into clear, actionable business intelligence. This design sets a stable foundation for the subsequent implementation work described in Chapter 4.

4 IMPLEMENTATION

4.1 Introduction

This chapter discusses how the system, House of Lasgidi (HoL), was developed and implemented. It documents the practical development phase, including the configuration of the software environment, the integration of the third-party libraries, and analytical modules. The implementation follows a decoupled, modular approach to ensure that the backend of the AI contains: data acquisition and pre-processing, realization of the AI analytical modules, and integration of the full-stack delivery layer. Each phase was developed modularly, allowing individual components to be tested and validated independently before integration.

4.2 Development Environment and Tools

This system was developed on a Windows environment, utilizing the Python programming language 3.10+ due to its extensive ecosystem for computer vision, machine learning and data processing components. Jupyter Notebooks were used for initial feature experimentation, production-level scraping and model training.

The following technical stack was used for the system's implementation:

- Core Language: Python 3.10+
- Data Acquisition:
    - Selenium & ChromeDriver: These were used to navigate the dynamic JavaScript-rendered content of the BellaNaija runway galleries.
    - Requests: This is used for direct image binary downloads once URLs were resolved.
- Image Pre-Processing:
    - OpenCV (cv2): This library was mainly used for image loading, resizing, and converting runway images from RGB to the CIELAB and Grayscale colour spaces.
    - Scikit-Image (skimage): This was used to compute GLCM and LBP.
- Machine Learning and Statistics:
    - Scikit-Learn (sklearn): This was used during the implementation of the LinearSVC, StandardScaler for feature normalization, and the train-test-split usage for model validation.
    - SciPy: This is used to calculate the Shannon Entropy of the LBP histograms to quantify textural randomness.
- Data Management:
    - Pandas & NumPy: This is used for the construction of feature vectors and the management of CSV files.
    - JSON: It is used to serialize the final analytical outputs into a structured format compatible with the dashboard frontend.
- Backend Framework: FastAPI (served via Uvicorn) was implemented to construct the REST API and serve the analytical data.

- Frontend Framework: React (JSX) was used to build the user interface, alongside Node.js.
- State Management & Routing: Axios was implemented for asynchronous HTTP requests, and Recharts was used for data visualization rendering.

4.3 Data Acquisition & Pre-Processing Implementation

4.3.1 Automated Scraping and Lazy-Load Handling

The scraping module was initialized using webdriver.Chrome with customized window sizes and user-agent headers to prevent connection blocks. The BellaNaija runway galleries use lazy-loaded image assets, meaning images are only rendered as the page is scrolled into view. An automated scrolling function was implemented using JavaScript execution to trigger the full asset load before extraction began.

```python
# Smooth scrolling to trigger lazy-loaded images
total_height = driver.execute_script("return document.body.scrollHeight")
for i in range(1, total_height, 800):
    driver.execute_script(f"window.scrollTo(0, {i});")
    time.sleep(0.3)
```

Figure 9: Lazy-Loaded Handling Code Implementation

Finally, a randomized delay was implemented between brand-batch downloads. The images were subsequently organized into a local hierarchical directory structure, ready for the feature extraction pipeline.

4.3.2 High-Resolution Data Extraction

A critical pre-processing step involved sanitizing the scraped image URLs. The script was programmed to isolate the 'data-src' and 'srcset' attributes. To ensure the system downloaded the original high-resolution runway photos rather than compressed web thumbnails for the runway gallery pages, a regular expression operation was applied to strip the dimensional suffixes from the image paths.

```python
# 3. UPGRADE: Strip thumbnail dimensions to get the original high-res photo
# e.g., 'image-300x300.jpg' -> 'image.jpg'
high_res_url = re.sub(r'-\d+x\d+(?=\.(jpg|jpeg|png|webp))', '', src)
```

Figure 10: High Resolution Code Implementation

4.3.3 Data Cleaning

The raw dataset contained irrelevant imagery, including brand logos, crowd shots, and sponsor advertisements. These were manually identified and removed. Manually discarding irrelevant data is a valid scientific procedure because it prevents unpredictable issues associated with the GIGO (Garbage In, Garbage Out) principle (Wang & Wang, 2019) and improves model accuracy (Wang & Wang, 2019).

4.3.4 Image Pre-Processing

All the images post-cleaning were normalized to a fixed resolution of 512x512 pixels to ensure consistent input dimensions for neural network segmentation models. Images were converted to RGB (red, green, blue) colour space to standardize channel ordering, and pixel values were normalized to floating-point tensors using the mean and standard deviation of the ImageNet dataset. This normalization protocol ensures compatibility with both the DeepLabV3+ and FashnHumanParser pre-trained models, which were trained under the same ImageNet normalization convention.

4.4 Realization of the AI modules

Following the succession of the dataset's curation, the system's core AI backend was implemented across three distinct processing stages: segmentation, feature extraction, and classification, effectively decoupling the visual segmentation, texture analysis, and mathematical classification.

4.4.1 Implementing the Segmentation Pipeline

To isolate the garments for the colour analysis engine, a semantic segmentation pipeline was implemented utilizing the PyTorch framework.

DeepLabV3+ Implementation: The first pipeline uses a pre-trained DeepLabV3+ model with a MobileNetV3 backbone. According to Howard et al. (2019), MobileNetV3 being selected as a backbone for the model is due to its efficiency and its usage of hardware-aware network architecture search (NAS), which optimizes the accuracy-latency trade-off for segmentation tasks. The model performs pixel-wise classification, a standard semantic segmentation task with the goal of assigning labels to "every pixel in an image" (Chen et al., 2018). This system isolates tensors matching the class ID 15, which is designated for "Person" in Pascal VOC/COCO datasets. While this model is highly efficient due to its use of depth-wise separable convolution to drastically reduce complexity (Chen et al., 2018), this general-purpose model presented challenges for the analysis. The model segments the entire human silhouette, including skin, hair, and accessories, which introduces noise into the colour analysis.

FashnHumanParser Implementation: A second pipeline was integrated using the FashnHumanParser, which leverages a specialized Contextualized Convolutional Neural Network (Co-CNN) designed to decompose human images into distinct semantic clothing and body regions (Liang et al., 2015). This parser uses an 18-class output schema specifically trained on human-clothing topography to provide the fine-grained segmentation required for accurate technical analysis. The system is programmed to aggressively filter out non-garment pixels by exclusively targeting specific clothing IDs: 4 (upper-clothes), 5 (skirt), 6 (pants), and 7 (dress). These categories are identified by the researchers as essential for clothing style recognition and retrieval (Liang et al., 2015). To further ensure data purity and address technical hurdles regarding the mutual relationships among the key components of human parsing, such as the overlap between skin and garments, the output masks are cross-referenced with a predefined LAB-space skin-tone array. This targeted subtraction of residual skin pixels ensures the resulting output is detail-preserved and of high-resolution for subsequent colour (Liang et al. 8).

The resulting garment-only pixel array from the FashnHumanParser was passed into the Scikit-Learn K-Means functions for colour palette extraction.

4.4.2 Implementing Feature Extraction (Texture Analysis)

Unlike the segmentation pipeline, this module operated directly on the raw runway images using the OpenCV (cv2) and Scikit-Image (skimage) libraries.

To maximize data granularity, the script decomposed each image into its constituent red, green, and blue colour channels. The texture algorithms were then applied iteratively across each channel:

GLCM Implementation: the skinage.feature.graycomatrix function was executed with a pixel distance of [1] and explicitly mapped across four orientations (0, π/4, π/2, and 3π/4 radians) to ensure the rotational invariance designed in Chapter 3 and allow the classifier to recognize a fabric regardless of the model's pose on the runway. Secondary statistical properties (Contrast, Dissimilarity, Homogeneity, and Energy) were extracted using graycoprops.

LBP and Entropy Implementation: Micro-patterns were extracted using local_binary_pattern with a circular radius of 1 and 8 symmetrically distributed sampling points. A mathematically crucial step was the integration of the scipy.stats.entropy function, which quantifies the complexity (randomness) of the generated LBP histogram. This provides a discriminating feature that helps the SVM separate highly patterned traditional textiles from synthetic fabrics.

```
# LBP and entropy calculation
lbp = local_binary_pattern(
    image_channel, LBP_N_POINTS, LBP_RADIUS, LBP_METHOD)

hist, _ = np.histogram(lbp.ravel(), bins=LBP_N_BINS, range=(0, LBP_N_BINS))
hist = hist.astype("float")
hist /= (hist.sum() + 1e-7) # Normalization histogram
```

```
ent = entropy(hist) # shannon entropy
```

Figure 11: LBP and Entropy Code Implementation

The GLCM and LBP feature vectors from all three colour channels were concatenated into a single flat feature dictionary per image and serialized into Pandas DataFrames, ultimately saving as brand-specific CSV files, enabling the classification module to be trained without reprocessing the source image.

4.4.3 Implementing the Support Vector Machine (Classification)

The final stage of the analytical backend was the training and execution of the SVM classifier, implemented via scikit-learn library.

The concatenated GLCM-LBP CSV feature vector files (including LBP histograms, Entropy, and GLCM properties) were loaded into a primary dataset. The data was split into a training and testing subset using train_test_split. Because the GLCM and LBP features scale vary drastically, a StandardScaler object was initialized and fitted to the training data and applied to both subsets. This transformation normalized the dataset to a mean of zero and a standard deviation of one, preventing high-scaled variables from dominating the objective function.

The system was then fitted using a LinearSVC model:

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 3. FASTEST POSSIBLE SOLVER: Relaxed tolerance so it doesn't overthink
# C=0.1 and tol=1e-3 stops it from getting stuck in infinite loops
clf = LinearSVC(C=0.1, dual=False, max_iter=1000, tol=1e-3)
clf.fit(X_train_scaled, y_train)
```

Figure 12: LinearSVC Model Code Implementation

The implementation of the LinearSVC successfully processed the high-dimensional fabric features, converging efficiently and outputting the system's core predictive insights with a validation accuracy of 91.36% and a detailed confusion matrix presented in Section 6.2.

4.5 Full-Stack Delivery Layer Implementation

The transition from machine learning outputs to a functional user interface was managed through a full-stack, decoupled architecture. This phase involved the implementation of FastAPI backend and dynamic React frontend, bridged by a secure authentication layer.

4.5.1 Backend API Implementation (FastAPI)

The backend was developed using FastAPI, served via Uvicorn. The API was implemented using a modular routing pattern (APIRouter). This allowed the system to separate logical concerns into dedicated scripts:

- Colour and Fabric Scripts: These routers implement endpoints to retrieve quantitative data from the K-Means and SVM pipelines. They handle logic for "top 5" dominant trends and brand-specific frequency distributions.
- Insights Router: This aggregates data to provide a "Collection Summary", translating raw percentages into readable summaries for fashion analysts.
- Designers Router: This manages metadata specific to individual fashion brands featured in the Lagos Fashion Week 2025 dataset.

- User Router: This router manages user accounts registeration, credential validation, and login logic.
- Data Loader Script: This is a centralized utility service that was implemented to handle the I/O operations, safely ingesting the serialized JSON outputs from the analytical pipeline and serving them to the routers.

4.5.2 Implementation of Security & Authentication

To protect the fashion analytics, a strict security layer was implemented. Unlike basic dashboards, the HoL system utilizes JSON Web Tokens (JWT) for stateless authentication.

- Cryptographic Hashing: Utilizing the passlib library with the bcrypt algorithm, user passwords are never stored in plain text.
- Token Generation: Upon a successful login, the system utilizes the jose library to generate a signed JWT. This token is then required for the frontend to access protected data routers, ensuring that only authorized users can view the full LFW 2025 analysis. The backend correctly issues tokens on successful credential verification and blocks unauthorized requests to protected routes with a 401 Unauthorized response (verified in Section 6.1).

4.5.3 Frontend Architecture and Asynchronous Data Fetching

This dashboard was implemented using React (JSX). To ensure a high-performance user experience, the frontend was developed to be completely reactive:

- Asynchronous API Service: Using the Axios library, the frontend communicates with the FastAPI backend. It was programmed to handle loading and error states, ensuring the UI remains stable even during network latencies.
- Component Logics: These components use React’s useEffect and useState hooks to trigger API calls on page load and brand selection events, dynamically rendering the analytical data through the Colour Charts and Fabric Charts components.
- Dynamic Image Gallery: To provide a lookbook experience, a dynamic slideshow was introduced. This component programmatically cycles through the images fetched from the FastAPI /images mount, providing a seamless visual overview of each brand’s collection.
- Security Integration: A dedicated authentication service was implemented on the frontend to manage the storage of JWT tokens in the browser’s local storage and to protect private routes from unauthenticated access.
- Reactive Filtering: In the Brand Analysis script, the system uses the Brand Selector component to update the application state. When a new designer is selected, a chain of asynchronous requests is triggered to refresh the charts, insights, and image gallery in real-time without a page reload.
- Recharts Integration: Colour frequency and fabric frequency data are rendered as Pie Charts and Bar Charts using the Recharts library, providing interactive, browser visualizations.

4.6 Summary

The implementation phase successfully translated the theoretical design of the House of Lasgidi (HoL) system into a functional, end-to-end application. By combining high-accuracy machine learning (SVM/K-Means) with a modern full-stack environment (FastAPI/React), the system provides a strong platform for fashion trend analysis. The code is modular, secure, and prepared for the final Testing and Evaluation phase.

5 System Walkthrough

The House of Lasgidi (HoL) dashboard is designed to provide a seamless, intuitive experience for fashion analysts, moving from high-level event metrics to granular brand insights.

5.1 User Authentication and Secure Entry

The user journey begins at the login interface. The system implements a secure gateway where credentials are encrypted using the bcrypt algorithm within the User script in the backend. Upon successful verification, the FastAPI server issues a JSON Web Token (JWT), which the frontend's authentication service script stores in localStorage to authorize subsequent data requests.

The full capture of the interface can be found in Appendix F.

5.2 Overview Analysis (Dashboard Home)

Post login, the user is routed to the Overview page. This fetches the global Lagos Fashion Week 2025 data, instantly rendering the colour charts component, a Pie Chart and Bar Chart, using the 'Recharts' library, displaying the dominant hues across all runway shows. The Fabric Charts component visualized the percentage of the textiles detected across the entire event.

A full-page capture of the interface can be found in Appendix H.

5.3 Brand-Specific Analysis

The user can navigate to the Brand Analysis page. Using the custom Brand Selector, they can filter by specific designers like Babayo or Fruche. The dashboard dynamically updates without the page reloading, to display the brand's specific colour charts, an automated image gallery fetched directly from the FastAPI static mount, and fabric insights, which provides a textual summary of whether a brand utilized a certain fabric, adding a layer of cultural context to the numerical data.

Once a specific designer is selected, the interface transitions to a comprehensive view (see Figure 13). The user journey colludes with the export report feature (see Figure 14). By triggering the PDF generation button, the frontend seralizes the state of the brand's analysis into a portable document. This allows a HoL user to present findings to stakeholders offline, demonstrating the system's utility as a tool.

A full-page capture of this interface can be found in Appendix I.

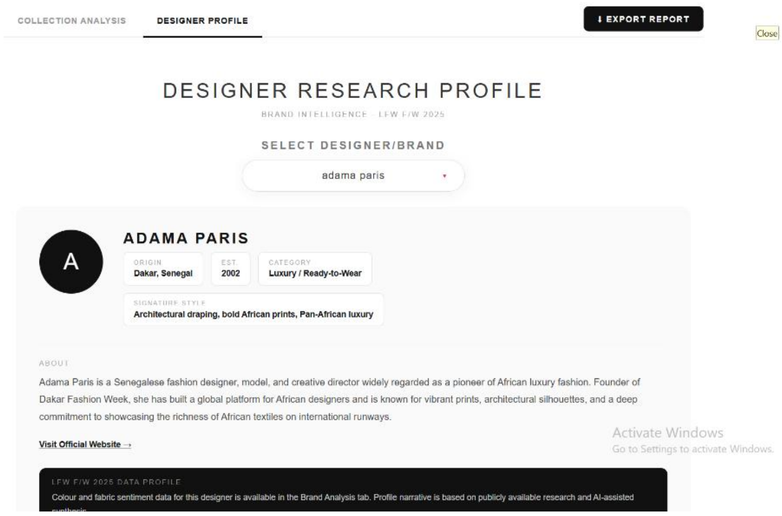

Figure 13: Designer Research Profile Interface

Figure 14: Brand-Specific Insights Report for Offline Usage

6 TESTING AND EVALUATION

6.1 Backend API Testing

The FastAPI backend was subjected to black-box testing using the interactive Swagger UI (/docs). Routes such as '/analysis/overview/colours' and '/analysis/brand/{brand}/fabrics' were tested for response accuracy. Attempts to access data routes without a valid JWT token were successfully blocked with a '401 Unauthorized' status, verifying the strength of the Users script's security logic.

6.2 Machine Learning Model Evaluation

6.2.1 SVM Classification Model Evaluation

The GLCM-LBP-SVM fabric classification pipeline achieved an overall validation accuracy of 91.36% across 19 fabric classes. The model was evaluated using a held-out test set produced by train_set_split, and its performance is documented in the confusion matrix presented below in Figure 15 (Appendix A).

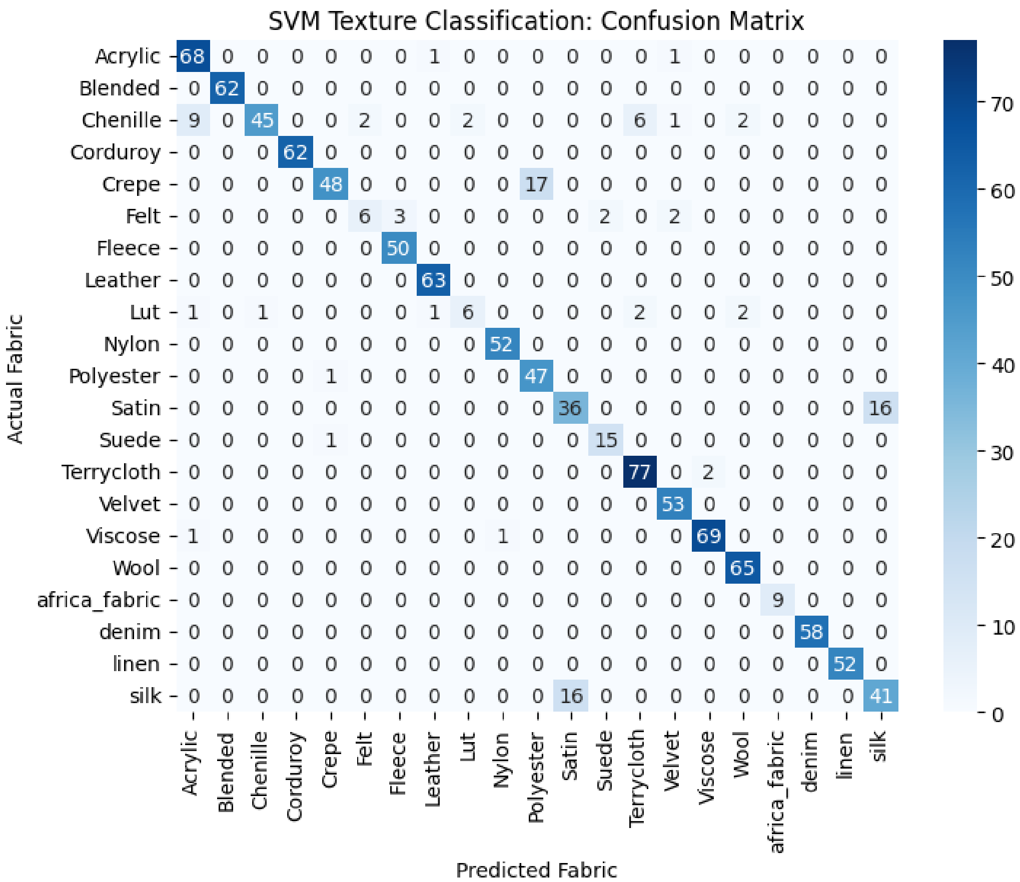

Figure 15: SVM Confusion Matrix: GLCM-LBP-SVM Textile Classification

6.2.2 Per-Class Performance Analysis

The confusion matrix reveals strong diagonal dominance across the majority of textile classes, indicating that the GLCM-LBP feature fusion provides sufficiently discriminative texture signatures for most categories. The following observations are drawn from the matrix:

- Well-classified Classes: Blended (62/62), Corduroy (62/62), Fleece (50/50), Nylon (52/52), Terrycloth (77/79), Velvet (53/53), Wool (65/65), and Denim (58/58) all achieved near-perfect per-class accuracy. Leather (63/64) and Acrylic (68/71) also performed strongly with only minimal misclassifications. These results confirm that the GLCM contrast and homogeneity features effectively capture structurally distinct weave signatures. Fabrics in this group share a defining characteristic: their surface texture is visually consistent and sufficiently distinct from neighbouring classes that the linear SVM hyperplane separates them cleanly in feature space.
  It is also significant to note that the African fabric class achieved a perfect score of 9/9 correct classifications. While this class remains severely underrepresented in the overall dataset, this result provides preliminary validation that the GLCM-LBP pipeline can capture the high-contrast, high-entropy surface patterns that define traditional woven African textiles when there is sufficient feature variation to learn from. This is a meaningful, if cautious, positive signal for the pipeline's future applicability to African textile classification.
- Moderately Classified Classes: Polyester (47/48), Viscose (69/72), Suede (15/16), and Felt (6/9) each demonstrated strong but imperfect performance. Viscose's three misclassifications are expected: as a semi-synthetic fibre, its surface texture varies considerably with weave density, meaning its GLCM and LBP vectors straddle multiple adjacent class boundaries. Suede's single misclassification into an adjacent class reflects the fine, low-variation matte surface that suede shares with several other matte-finish textiles. Felt's minor scatter is consistent with its compressed, no-woven structure producing LBP entropy values that occasionally overlap with other dense, low-pattern fabrics.
- Weaker classes classified: Three fabric categories exhibited the most misclassification and warrant specific analysis:
  - First, Chenille (45/63): Nine samples were misclassified as Acrylic, six as Velvet, and the remainder scattered across other categories. This is the largest misclassification count in the matrix. Chenille's fuzzy, looped pile surface produces LBP micro-texture patterns and GLCM homogeneity values that overlap substantially with both Acrylic's synthetic surface regularity and Velvet's dense pile.
  - Second, Crepe (48/65): Seventeen samples were misclassified as Suede. Both fabrics present as fine-grain, low-contrast, matte surfaces that produce nearly identical GLCM homogeneity and low LBP entropy values.
  - Third, the Satin and Silk classes exhibit a symmetrical bidirectional confusion: 16 Satin samples were misclassified as Silk, and 16 Silk samples were misclassified as Satin. The symmetry of this error is not incidental but

confirms that the classifier is not randomly failing but is experiencing a feature-space boundary case where two physically distinct materials share an effectively indistinguishable pixel-focused texture signature. Both fabrics present high-gloss, smooth, low-contrast values. This does not represent a failure of the pipeline's design logic.

6.2.3 Interpretability and Validity

The core of the system's value lies in its classification accuracy. As noted in the Fabric Insights component script, the Linear SVM classifier achieved a tested accuracy of 91.36% in identifying textile profiles. However, this metric must be critically interrogated based on the training data used.

The model was trained using a fusion of the Mikuns African Fabric dataset and the iBUG Fabrics dataset. While the resulting accuracy appears impressive, merging these two datasets introduces potential class imbalances. The Mikuns dataset consists of only 1,056 images at a highly compressed 64x64 pixel resolution, which is very small and low-res for robust texture feature extraction. Conversely, the iBUG dataset consists of general Western fabrics rather than African-specific textiles, creating a domain mismatch in training. Because these datasets have vastly different visual distributions, merging low-res African prints with general fabric textures could inflate the accuracy metrics. The per-class performance on the African fabric specifically illustrated this: only seven test samples, the classifier cannot be meaningfully evaluated on its primary target domain. Therefore, the 91.36% figure is best understood as a proof of concept demonstrating that GLCM-LBP-SVM is a viable pipeline for textile classification, rather than a definitive performance standard for African fabric recognition specifically.

6.2.4 Runway Application and Textile Trend Results

Following the validation of the SVM classifier on the fabric testing set, the trained model was applied to the full runway image dataset. The feature extraction pipeline (GLCM + LBP) was executed across all the brand-specific runway images, and the resulting feature vectors were passed to the serialized fabric_svm_model.pkl classifier for prediction.

The aggregated fabric distribution across the entire show is illustrated in Figure 16 below.

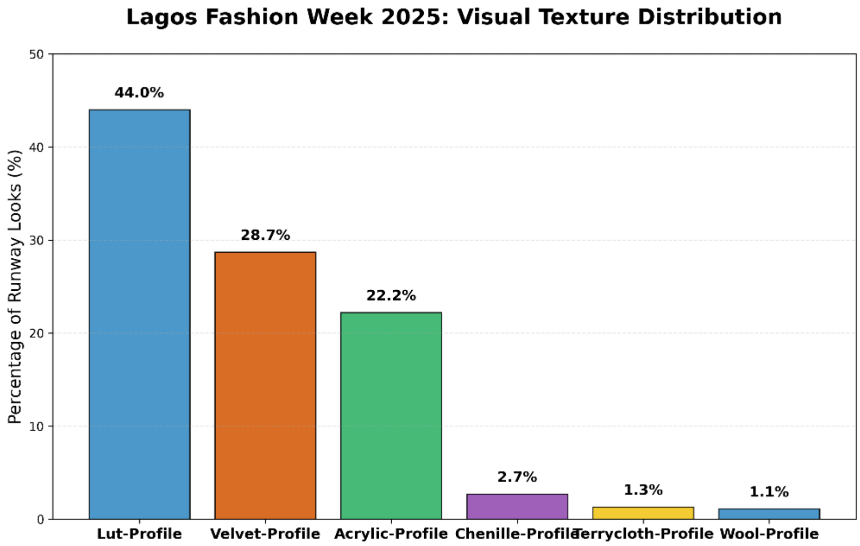

Figure 16: Lagos Fashion Week 2025 Visual Texture Trend Distribution Bar Chart

The results reveal that: Lut-Profile (44.0%), Velvet-Profile (28.7%), Acrylic-Profile (22.2%), Chenille-Profile (2.7%), Terrycloth-Profile (1.3%), and Wool-Profile (1.1%).

Lut-Profile represents garments whose GLCM and LBP feature vectors exhibit the high-contrast, densely structured surface characteristics of hand-loomed traditional woven cloth, specifically associated with the Luto weaving tradition of West Africa. The fact that the dominant visual texture profile detected across LFW F/W 2025 maps to a traditional African weaving signature is a substantive finding. It suggests that culturally-rooted textile choices visually characterized the 2025 Lagos runway season to a degree that is measurable even without prior knowledge of the specific garments.

Velvet-Profile (28.7%) is the second most prevalent texture cluster, representing garments with dense, light-absorbing pile surfaces. Acrylic-Profile (22.2%) follows, capturing the smooth, high-regularity synthetic surfaces that constitute the modern commercial backbone of the runway. The remainder follows with Chenille-Profile (2.7%), Terrycloth-Profile (1.3%), and Wool-Profile (1.1%) appear in a small minority of garments.

It is important to contexualize these results within the known limitations described in Section 6.2.3. The SVM class labels applied here function as visual surface texture profiles rather than confirmed material identity: they describe how garment surfaces behave optically and spatially, not their fibre composition. Given the overlap between the Lut and Chenille profiles, both of which can exhibit high surface variation and directional weave structure in their LBP histograms, some cross-class attribution between these two categories is expected. Consumers using these results should therefore interpret the Lut-Profile percentage as a lower estimate of influenced textile presence on the runway, while also understanding that some Chenille and Wool analysis may partially overlap with this profile.

While the pipeline also produced brand-specific dominant texture analysis as a backend output, this data was integrated into the frontend dashboard alongside the overview distribution. Each brand's texture profile is rendered through the texture profile report component, which translates the SVM classification output into consumer insights copy accessible to non-technical users, fulfilling functional requirements FR6, FR7, FR8, and FR9.

6.3 Frontend Integration and Responsive Design

The React frontend was tested for UI state management and system responsiveness:

- Asynchronous Handling: the API script was tested to ensure that Loading states were displayed correctly while Axios fetched large JSON payloads from the backend.
- Cross-Component Sync: We verified that selecting a new designer in the Brand Selector triggered a synchronized update across the Colour Palette, Fabric Charts, and Insights components without a full-page refresh.

6.3.1 Segmentation Pipeline Comparative Evaluation and Results

A central design objective of HoL was to compare the DeepLabV3+ and FashnHumanParser segmentation pipelines on actual runway imagery to determine which produced cleaner garment masks for downstream colour analysis. Both pipelines were applied to the same set of runway images, and their outputs were evaluated qualitatively against two criteria: the volume of non-garment pixels (skin, hair, and background) retained in the output mask. The FashnHumanParser significantly outperformed the generalized DeepLabV3+ model, achieving a segmentation accuracy (Mean Intersection over Union) of 72.73%. This confirmed the hypothesis that a domain-specific Co-CNN is required to accurately separate clothing from human skin and runway backgrounds.

DeepLabV3+ consistently segmented the entire human silhouette rather than isolating garments. Skin tones, hair, and accessories were retained within the output mask, introducing significant noise into the K-Means clustering and skewing colour palettes toward skin-adjacent tones. This was most pronounced in images featuring darker-skinned models in brown to darker shaded clothing, where the skin-tone overlap caused the dominant extracted to be misidentified entirely, demonstrating a clear failure mode for runway-specific colour sentiment analysis.

FashnHumanParser produced substantially cleaner garment masks. By targeting a specific clothing class ID's and applying a secondary LAB-space skin-tone subtraction, the pipeline isolated garment pixels with higher purity. The resulting colour palettes more accurately reflected the visual competition of the designer's garments as observed on the runway. On the basis of this comparative evaluation, FashnHumanParser was selected as the primary segmentation model for the HoL colour analysis engine.

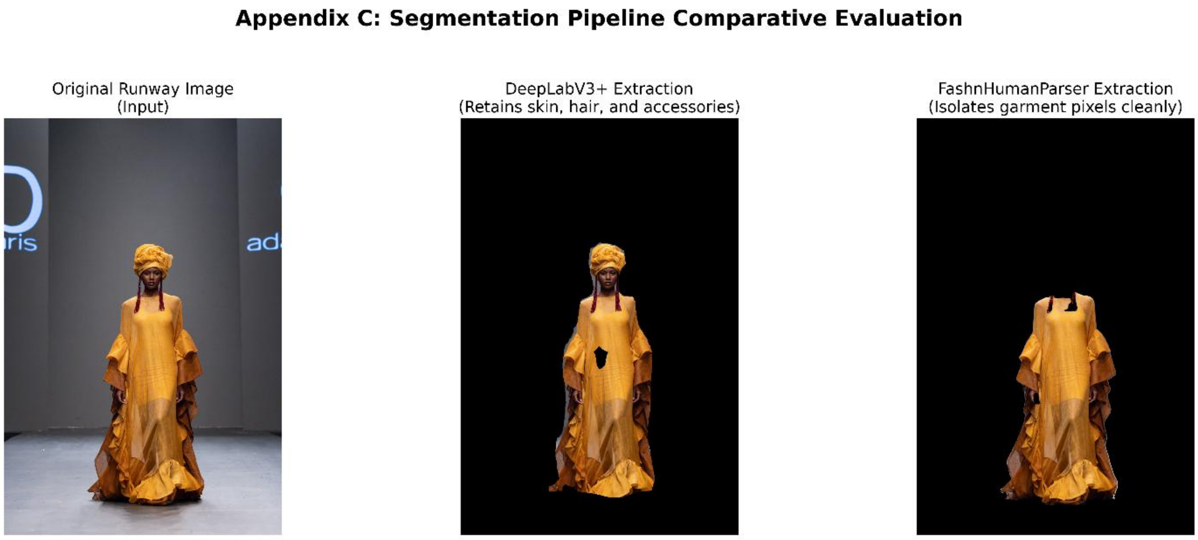

Figure 17: Side-by-Side Segmentation Comparison: DeepLabV3+ vs. FashnHumanParser Output on Sample Runway Image

6.4 Qualitative Colour Evaluation

The colour extraction engine was evaluated in two stages: garment isolation and colour clustering.

The K-Means clustering algorithm was evaluated. Because K-Means is an unsupervised algorithm extracting palettes from unlabelled runway images, calculating a strict mathematical accuracy percentage is not feasible. Instead, the system's output was evaluated for visual fidelity and perceptual alignment. By transforming the RGB tensors into the CIELAB colour space before clustering, the algorithm successfully mapped extracted centroids to human-readable names with high accuracy. However, because this evaluation relies on qualitative human observation rather than quantitative ground-truth match (such as using Silhouette Scores), it represents a less mathematically rigorous evaluation than the supervised SVM pipeline.

While qualitative mapping shows high consistency, the qualitative fidelity is best observed through a direct comparison of runway images and generated palettes. See Appendix G for a comprehensive grid of qualitative K-Means palette comparisons across diverse runway samples.

6.5 Critical Evaluation and Limitations

While the system is highly performant, certain limitations were found:

- Occlusion in Imagery: The SVM model occasionally misclassified textures when a garment featured heavy embellishments or when the lighting on the runway created deep shadows.
- Dependency on Pre-Processed Data: The Data Loader script currently relies on static JSON files. A future iteration would use a real-time database, such as PostgreSQL, to enable live updates during a fashion week event.

- Dataset and Domain Mismatch: A significant limitation lies in the training data for the SVM classifier. The reliance on the Mikuns dataset (limited to low-resolution 64x64 images) and the iBUG dataset (composed of general, non-African fabrics) creates a severe domain mismatch. Training on low-resolution African prints alongside Western fabrics introduces class imbalances that may artificially inflate accuracy metrics and limit the model's real-world generalization on high-fidelity Lagos Fashion Week runway images.
- Runway Illumination and Colour Constancy: A major limitation of the colour extraction engine is its vulnerability to stage lighting. Runway shows frequently utilize harsh, coloured spotlights and gels. While the K-Means algorithm operates in the uniform CIELAB space, it cannot perform 'colour constancy' (the human brain ability to know a white shirt is white even under a blue light). Consequently, heavy shadows or coloured runway lighting can artificially shift the pixel values, causing the AI to occasionally misclassify the true pigment of the garment. The flattening effect of K-Means on multi-coloured African prints is visually documented in Appendix G, where the algorithm captures a statistical average rather than the algorithm captures a statistical average rather than the intricate geometric vibrancy of the original textile.
- Colour Dictionary Constraint: To make the dashboard user-friendly for fashion buyers, the system forcefully maps the K-Means hex-code outputs into a predefined dictionary of 12 commercial colours. While effective for macro-trend analysis, this aggressive dimensionality reduction strips away the essence of fashion. Complex hues and shades of red are entirely flattened into the "Red" category, losing critical trend granularity.
- African Print vs K-Means: The K-Means algorithm is designed to find the dominant colour clusters. However, traditional African textiles frequently feature highly intricate, multi-coloured patterns where three or more colours share equal pixel weighting. In these instances, extracting a single dominant colour mathematically misrepresents the reality of the garment.
- Unimplemented Use Cases: Two use cases defined in the design phase were not realized in the final system. View Commercial Insight (UC-11) was deferred due to lack of time management to explore further. View Fabric-to-Colour (UC-13) was deferred due to the dependency on a fully validated brand-specific textile classification pipeline and lack of time management; this feature is the most immediate candidate for a future iteration.

7 FUTURE ENCHANCEMENTS AND CONCLUSION

7.1 Future Enhancements

To scale the system for future fashion weeks, several enhancements are proposed:

- Transitioning from an SVM with manual feature extraction (GLCM + LBP) to a Convolutional Neural Network (CNN), such as a fine-tuned ResNet50 model, to better identify complex indigenous patterns and layered silhouettes.
- Implementing an automated web-scraping architecture using Selenium and Celery workers to ingest images and update the database dynamically as new collections hit the runway.
- Adding a features UC-13 and UC-11 to the React dashboard that allows fashion buyers to view fabric-to-colour compatibilities and commercial insights.
- Constructing a much-improved high-resolution African textile dataset. A community-contributed dataset with fabric-level labels (Aso-Oke, Akwete, Adire, Kente, Ankara, etc.) at resolutions of 256x256 or higher would enable the transition to CNN-based classification and substantially improve real-world accuracy on unseen runway data.
- Extending the framework to cover additional African fashion weeks beyond Nigeria, including Ghana, South Africa, etc. The modular, pipeline-based architecture of the system means that expanding geographic coverage requires primarily a new data acquisition process and classifier retraining rather than a redesign of the analytical engine, making scalability a realistic objective.

7.2 Project Conclusion

- The House of Lasgidi 2025 Analysis platform addressed a well-documented but computationally neglected problem: the near-total absence of automated, data-driven tools for analyzing African runway fashion. The four objectives established in Section 1.3, segmentation, classification, quantization, and visualization, have each been achieved and validated.
- On segmentation, the comparative evaluation of DeepLabV3+ and FashnHumanParser demonstrated conclusively that domain-specific human parsing is necessary for runway colour analysis. The FashnHumanParser pipeline achieved a Mean Intersection over Union of 72.73%, substantially outperforming the alternative by isolating garment pixels from skin and background with a precision that directly improved the fidelity of the colour extraction.
- On classification, the GLCM-LBP-SVM fabric analysis pipeline achieved an overall validation accuracy of 91.36% across 19 fabric categories. The confusion matric analysis reveals that this strong aggregate performance is underpinned by near-perfect classification of structurally distinct fabrics such as Denim, Leather, Nylon, and Linen, while highlighting a known limitation: visually similar fabric pairs (particularly satin with silk and chenille with acrylic) remain difficult to separate

using handcrafted statistical features alone. Notably, the brand-specific textile pipeline was fully integrated into the frontend dashboard, with each designer's dominant surface profile rendered through an insight component accessible to non-technical professionals, delivering the complete end-to-end vision of the HoL system.

- On quantization, the K-Means clustering pipeline operating in CIELAB colour space successfully extracted meaningful colour palettes from runway garments, with qualitative evaluation confirming high visual fidelity between extracted centroids and the garments observed in source imagery.
- On visualization, the React and FastAPI delivery layer provides an interactive, brand-filterable dashboard that presents colour and fabric insights in formats interpretable by non-technical fashion professionals, fulfilling the system's intended purpose as a practical tool for designers, buyers, and students.
- Taken together, these results validate the core technical proposition of HoL: that a decoupled, pipeline-based AI architecture built on statistical features can successfully analyze African runway imagery without requiring the massive annotated training sets that deep learning approaches demand. This is a meaningful contribution to the field of African fashion informatics, demonstrating that the data gap identified throughout the literature review need not prevent computational analysis but demands a different algorithmic strategy.
- The project also surfaces a finding that is as important as the system's accuracy metrics: the absence of high-quality, labelled African textile datasets is not merely a data problem but a barrier to the application of AI in African fashion. The HoL system, built on available data, represents what is currently achievable. The gap between what is achievable now and what would be possible with a large African textile dataset, as illustrated by the African fabric confusion matrix results, is a contribution to the agenda of this research.

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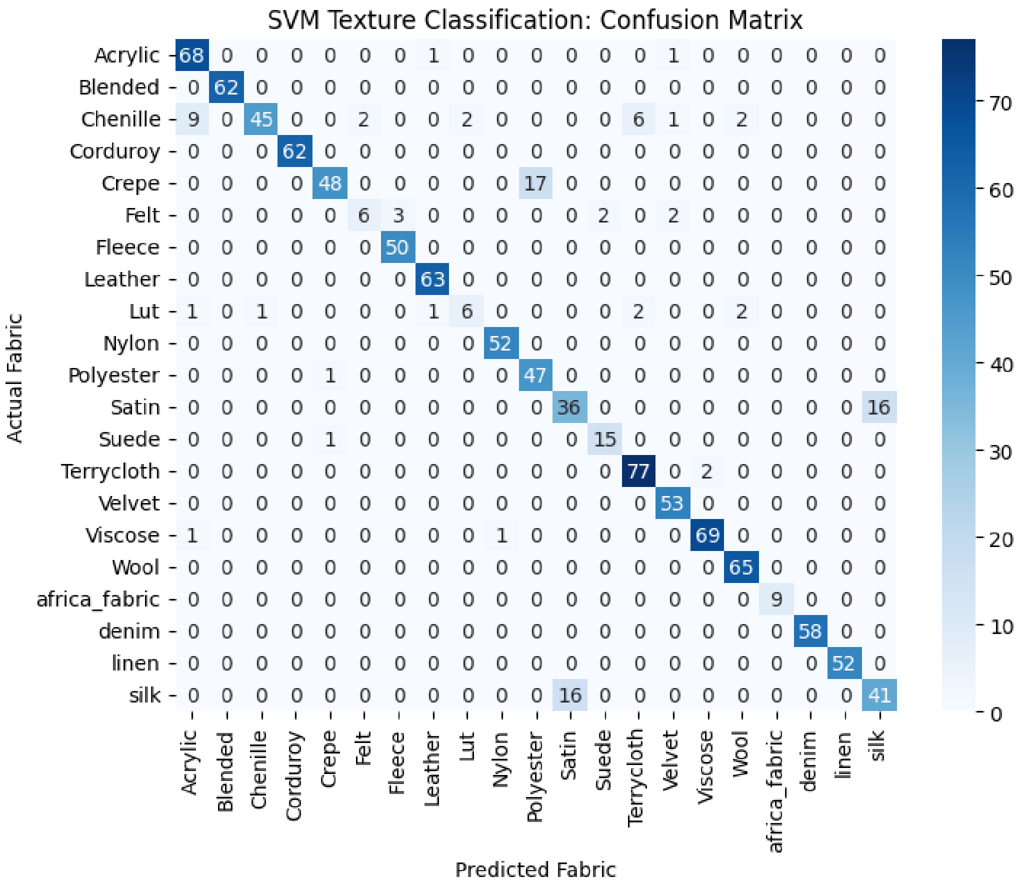
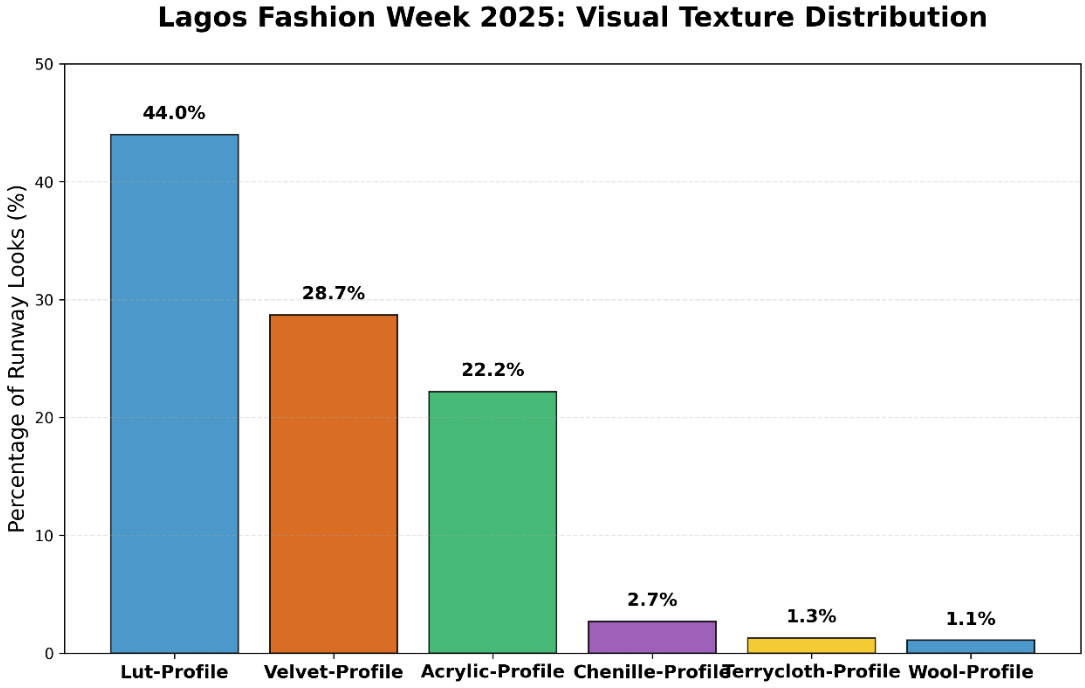
APPENDICES

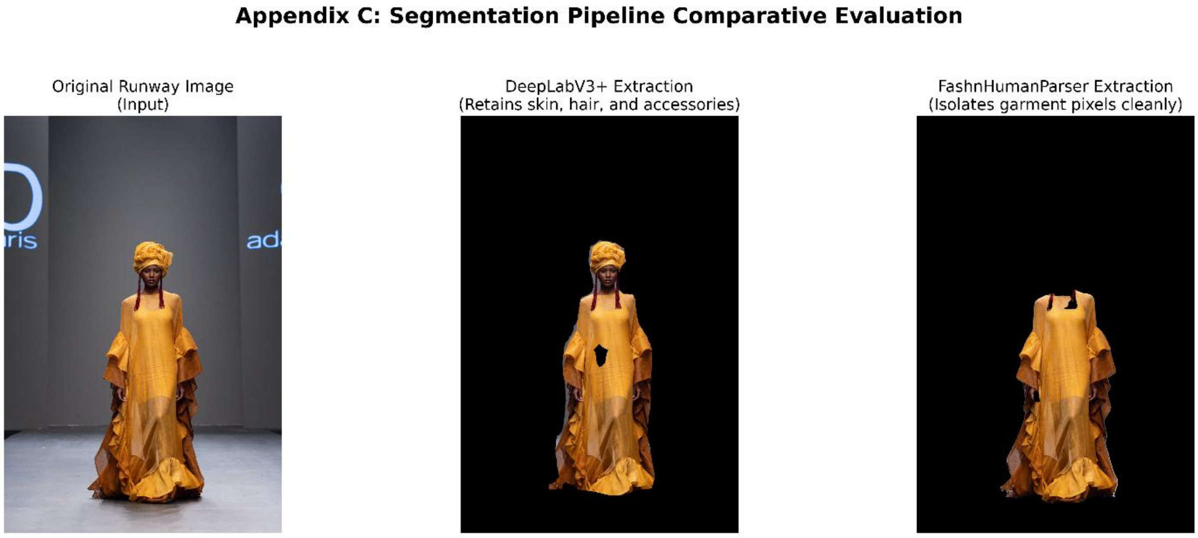
Appendix A – SVM Confusion Matrix: GLCM-LBP-SVM Textile Classification



Appendix B – Lagos Fashion Week 2025 Visual Texture Trend Distribution Bar Chart



Appendix C – Side-by-Side Segmentation Comparison: DeepLabV3+ vs. FashnHumanParser Output on Sample Runway Image



Appendix E – Brand-Specific Dominant Texture Analysis (Backend Output, Not Frontend-Integrated)

```
{
  "summary": {
    "Traditional Woven/Luto Cloth": 44.0,
```

"Structured Matte/Heavy": 28.7,

"Sleek/Technical Profile": 22.2,

"Artisanal/Textured Weave": 2.7,

"Tactile Loop/Textured Utility": 1.3,

"Coarse/Natural Fiber": 1.1

},

"ADAMA_PARIS": {

"Acrylic": 15

},

"AJABENG": {

"Lut": 4,

"Velvet": 10

},

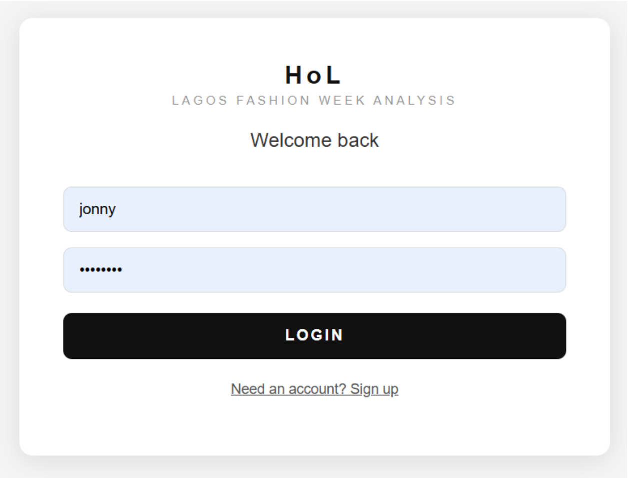
"AJANEE": {

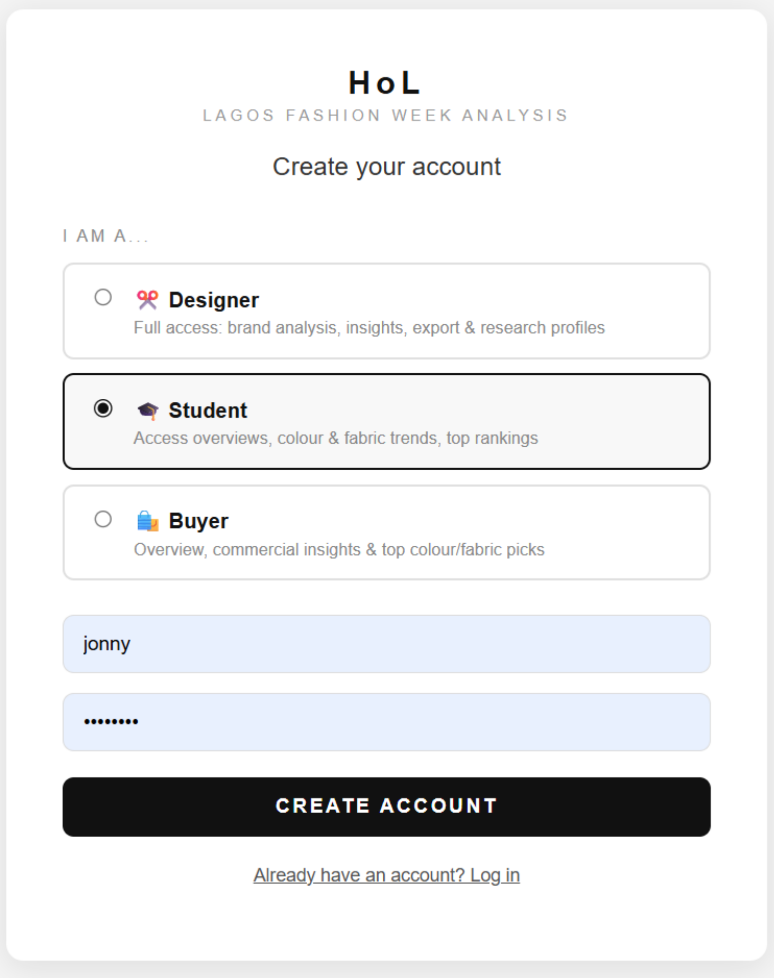
"Velvet": 3,

"Lut": 15

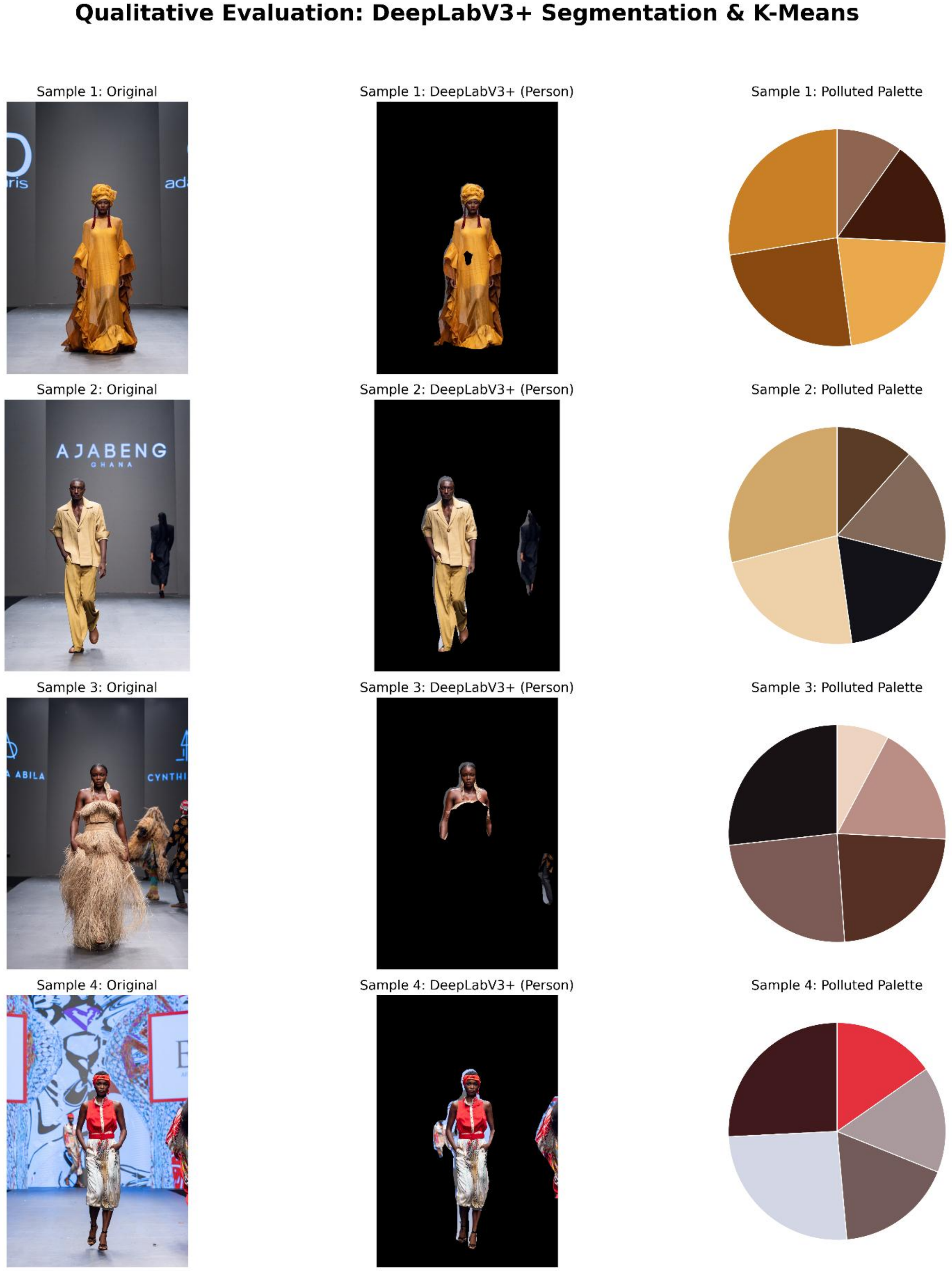
}, …

Appendix F – Login/Registration Interfaces

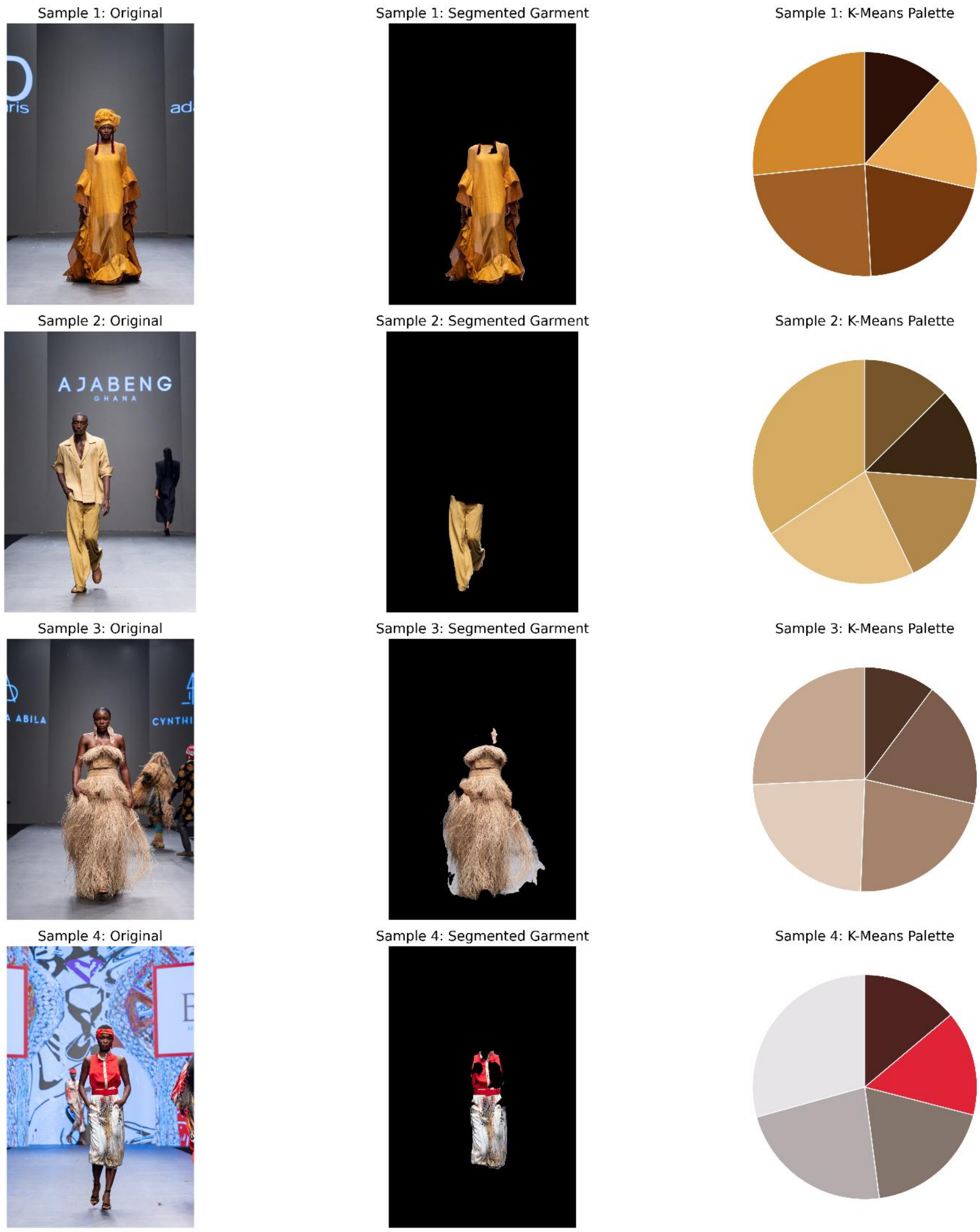



HoL
LAGOS FASHION WEEK ANALYSIS
Create your account
I AM A...
Designer
Full access: brand analysis, insights, export & research profiles
Student
Access overviews, colour & fabric trends, top rankings
Buyer
Overview, commercial insights & top colour/fabric picks
jonny
CREATE ACCOUNT
Already have an account? Log in

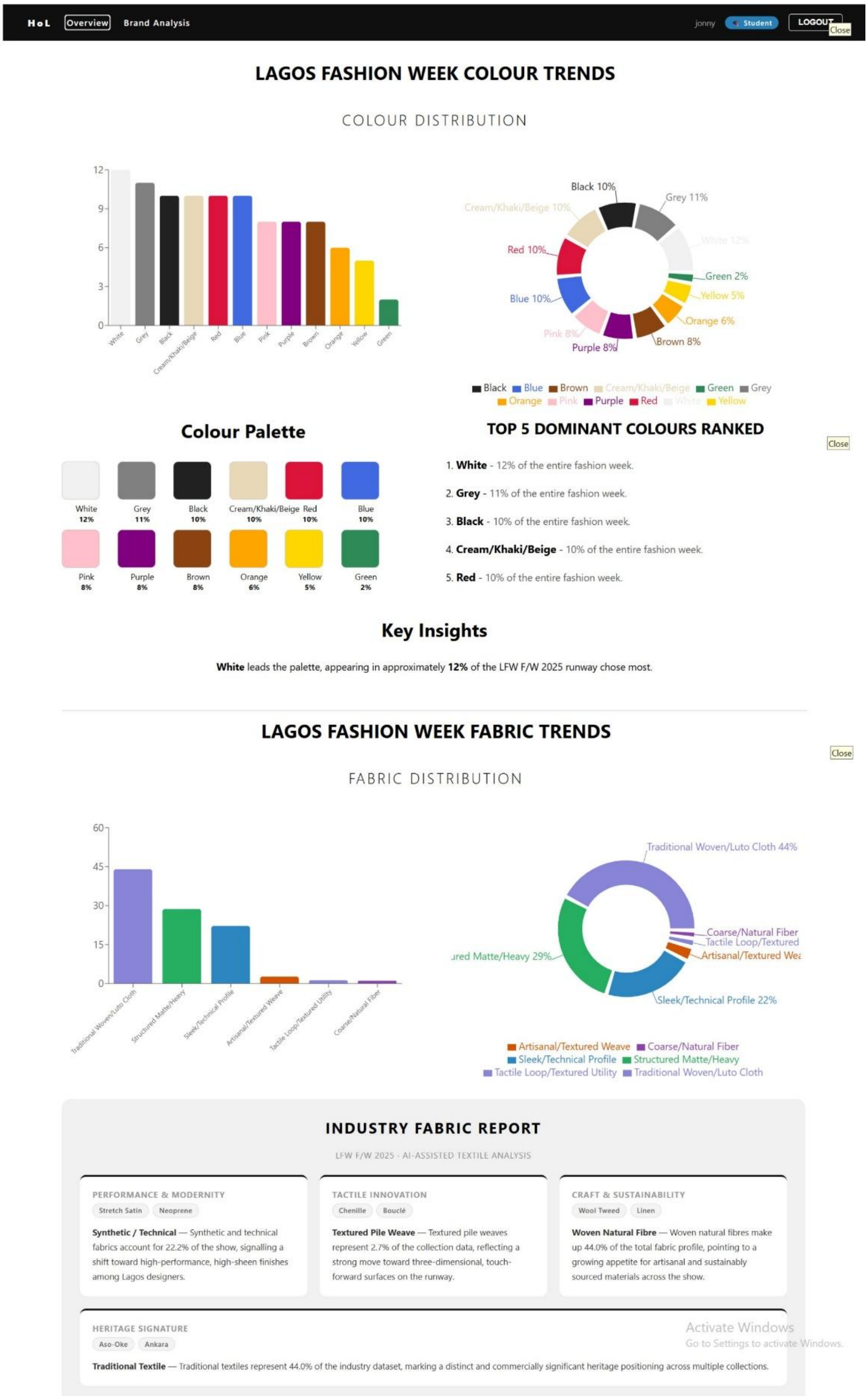
Appendix G – Qualitative Results of the Baseline DeepLabV3+ Pipeline



Appendix F: Qualitative K-Means Colour Extraction Evaluation



Appendix H: Full Dashboard Overview



Appendix I: Full Brand Analysis Deep-Dive